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THE DEFINITIVE GUIDE TO WHEY PROTEIN

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This guide is based on scientific studies, but individual results do vary. If you engage in any activity or take any product mentioned herein, you do so of your own free will, and you knowingly and voluntarily accept the risks.

Preface

Alex here. Hey guys and gals, you may know me as one of the researchers at Examine.com. I've been interested in nutrition for many years now, and have been involved in sports for most of my life. Supplements are therefore no stranger to me.

If you walk into any supplement store, or even your run-of-the-mill supermarket, you're probably going to see some whey protein powders on the shelves. You'll probably see a lot, actually, all claiming they're better than their neighbors.

Turn to the Internet, and the number of options becomes nearly infinite.

Which do you pick?

It's a simple question, yet one I could never find a good answer to, in any book, on any website. So I decided to write my own. With the support of the Examine.com team, I've spent months scouring the research. I soon came to realize that to answer my "simple" question, I needed to first answer many others, such as how manufacturing methods can impact quality and whether denaturation is something to worry about.

Further, I soon realized I couldn't even stop at the protein itself; I had to research all the ingredients commonly added to protein powders to improve them in some way … or simply justify a high price tag. I also looked at casein and amino acids that may be blended with whey protein or used in its place for a variety of reasons.

My goal with this guide is simple: to help you navigate the whey protein powder market. I want you to know what to look for in a product; what the red flags are; what questions to ask.

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FAQ

Is it true that protein-rich diets shorten your life?
There is evidence that diets low in calories and protein can increase lifespan, but it is weak. It derives mostly from studies in mice, whose lifespan and metabolism are very different from ours. Studies in monkeys do not link caloric restriction to increased lifespan.

This weak evidence in favor of a low protein intake must be contrasted against the far stronger data showing that, with aging, a low protein intake promotes frailty and muscle loss, increases the risk of premature death, and reduces quality of life.

Can't I just get my protein from whole foods?
You certainly can, but protein powders have several advantages: they're very low in carbs and fat, cheap (protein for protein), easy to carry, fast to prepare, easy to drink even when you're too full to eat, and easy to digest and absorb. Moreover, your immune system may benefit from whey protein's unique bioactive peptides.

Can a whey protein powder replace whole foods entirely?
At least one study demonstrated that replacing half your daily whole-food protein with whey protein didn't meaningfully affect physical fitness or body composition, but you still need whole foods for their vitamins, minerals, and essential fatty acids.

What's the difference between milk powder and whey protein powder?
The protein in cow's milk is only 20% whey protein (and 80% casein), but milk protein is good, that's not the problem. So what is the problem? Let's say you pick a skim milk powder, so your milk powder is nearly fat-free; it still has less protein (36%) than carbs (52%), and those carbs are lactose, a sugar. By contrast, a decent whey protein concentrate will be 80% protein, and its carbs will be in the low single digits.

What exactly is whey?
Whey comes from milk; it is most often a byproduct of cheesemaking. Whey's dry mass is 75% carbs (lactose), 13% protein, and 1% fat. So for those of you who asked if you could make your own whey protein at home, the answer is no, because separating the protein from the carbs requires heavy machinery.

Is whey the best protein source?
Whey protein is highly bioavailable and has an excellent amino acid profile.

- Animal-based proteins (such as whey) and plant-based protein powders are digested and absorbed with more than 90% efficiency, compared to 60–80% for the protein in plant-based whole foods.

- Whey protein is 52% EAAs and 13.6% leucine, whereas other animal-based proteins are roughly 40–45% EAAs and 7–8% leucine, and plant-based proteins are even lower.

Is isolate better than concentrate? What about hydrolysate?
There is no essential difference between whey concentrates and isolates: the latter just have a little less fat and carbs. Still, since the carbs are lactose, an isolate may suit you better if you're lactose intolerant.

In terms of their effects on strength and muscle mass, however, there doesn't appear to be meaningful differences between concentrates, isolates, and hydrolysates. Hydrolysates are "pre-digested" but neither more bioavailable nor faster digesting than concentrates and isolates. Moreover, hydrolysates are denatured proteins, which means your digestive enzymes may not be able to produce health-promoting peptides from them.

Does it matter how the whey protein was processed?
Some forms of processing can denature the protein, but it is still protein: your body will use it, including to build muscle. However, your digestive enzymes usually produce health-promoting peptides from the protein you ingest; they may not be able to do so if the protein is denatured.

What is cold-processed whey?
Cold-processed whey is whey that has been created without the application of heat. This term is doubly meaningless: all whey protein powders derive from pasteurized milk, and none of the filtration methods used to concentrate whey protein (i.e., to extract it from the whey) involve heat application. The only part of the whole process for which "cold processing" could possibly make sense is the transformation of the liquid whey protein into a powder: vacuum drying and freeze drying don't involve heat, whereas spray drying does.

Should my whey be from pasture-raised, grass-fed cows?
That's up to you. There are important environmental and ethical arguments to be made about how cows are raised, but from a purely nutritional standpoint, greater access to pasture doesn't appear to matter.

Any additives I should avoid?
Most additives aren't worth worrying about in the concentrations found in whey protein powders (if they're even there all). Only carrageenan and food dyes (artificial colorants) warrant caution.

Should I take a whey protein powder especially made for women?
Whey protein is whey protein. A whey protein powder can only be "made for women" through additional ingredients, such as added iron. It's up to you to decide if those added ingredients are worth your money. Of course, you'll find plenty of information on Examine.com to help you in your decision.

Is there anyone who shouldn't use whey?
Someone who is allergic to whey protein shouldn't use it. Otherwise, it is a personal choice.

Why can't I just use BCAAs?
Well, you can. But why when the benefits seen from whey protein are both larger and more consistent than the benefits seen from leucine, BCAA, or EAA supplementation? A complete, fast-digesting protein, such as whey, should be your first choice, but if for whatever reason a protein powder is not an option for you, then some isolated leucine, BCAAs, or EAAs may be useful, depending on your goals.

How much glutamine is there in whey protein?
None, but whey protein contains glutamic acid (a.k.a. glutamate), from which your body makes glutamine. Of course, glutamine is also sold as a supplement.

Should I take casein before bed?
If you haven't consumed enough protein during the day, then taking casein before bed can benefit you, but the same can be said of any other protein. Time of ingestion doesn't seem to matter: whether in the morning or near bedtime, a fast-digesting protein, such as whey protein, increases strength and muscle mass more than does a slow-digesting protein, such as micellar casein.

Should I combine whey with casein?
Milk protein (a 4:1 ratio of micellar casein to whey protein) and whey protein have similar effects on muscle protein synthesis, muscle mass, and strength. Likewise, milk protein, whey protein, and a 1:1 blend of whey protein and micellar casein have similar effects on muscle mass and strength. Hence, given proteins of similar quality, a blend of slow- and fast-digesting proteins won't benefit your muscles more than just a fast-digesting protein.

Is hydrolyzed collagen (collagen peptides) better than whey protein?
It depends on your health goal. Unlike other animal-based protein powders (whey, casein, egg; beef protein is most often collagen under another name), hydrolyzed collagen is not a complete protein.[1] Rich in glycine and proline but poor in BCAAs, it isn't a good primary source of protein, and is probably not the best muscle builder (though it has shown benefit in elderly women on a low-protein diet[2] and in elderly men[3]).

It is true, however, that collagen is the most common protein in your body: most of your skin, joints, and bones are made of collagen. Studies have shown that collagen protein benefits skin and joints, and there is mechanistic evidence that it can benefit bones too.

Of course, nothing prevents you from taking both whey protein and hydrolyzed collagen.

How much protein do you need?

As with most things in nutrition, there's no simple answer. Your optimal daily protein intake depends on various factors, notably your health goal, body composition, and level of physical activity. And even taking all this into account, you'll end up with a *starting* number, which you'll need to adjust through self-experimentation.

Table 1: Optimal daily protein intake in grams per kilogram of body weight (g/kg)

	Of healthy weight			Overweight or obese	Pregnant	Lactating
	Maintenance	Muscle gain	Fat loss			
Sedentary	1.2–1.8	n/a		1.2–1.5	1.66–1.77	>1.5
Active	1.4–2.2	1.4–3.3	2.3–3.1		unknown	unknown

From How much protein do you need per day? (published January 16, 2013; last updated June 5, 2019; accessed June 7, 2019)

It is easy to assume that getting more daily protein than the US Recommended Dietary Allowance (RDA) serves no purpose.[4] Despite their name, though, RDAs do not represent *optimal* intakes; rather, they represent the *minimum* needed by healthy, sedentary adults to avoid deficiency-related health issues (such as scurvy from not enough vitamin C).

In the case of protein, the 0.8 g/kg RDA represents the minimal amount a healthy, sedentary adult needs daily to prevent muscle wasting when total caloric intake is sufficient. This number has been challenged, however: 1.2 g/kg has been suggested as a better number[5,6,7,8] by studies that used the Indicator Amino Acid Oxidation (IAAO) method to overcome many of the shortcomings of the nitrogen-balance studies used to establish the RDA.[9]

For example, nitrogen-balance studies require that people eat experimental diets for weeks before measurements are taken. This provides ample time for the body to adapt to low protein intakes by downregulating processes that are not necessary for survival but are necessary for optimal health,[10] such as protein turnover and immune function. By contrast, the IAAO technique takes just 24 hours to determine protein requirements.

Of course, if you are not sedentary, if you exercise regularly through work or leisure, then you need even more protein. The American College of Sports Medicine, the Academy of Nutrition and Dietetics, and the Dietitians of Canada recommend 1.2–2.0 g/kg to optimize recovery from training and promote the growth and maintenance of lean mass when total caloric intake is sufficient.[11] This recommendation is similar to the 1.4–2.0 g/kg promoted by the International Society of Sports Nutrition (ISSN).[12]

Importantly, it may be better to aim for the higher end of the above ranges. According to the most comprehensive meta-analysis to date on the effects of protein supplementation on muscle mass and strength,[13] the average amount of protein required to maximize lean mass is about 1.6 g/kg, and some people need upward of 2.2 g/kg.

However, only 4 of the 49 included studies were conducted in people with resistance-training experience (the 45 others were in beginners). IAAO studies in athletes found different numbers: on training days, female athletes required 1.4–1.7 g/kg;[14, 15] the day following a regular training session, male endurance athletes required 2.1–2.7 g/kg;[16] two days after their last resistance-training session, amateur male bodybuilders required 1.7–2.2 g/kg.[17]

And this is when calories are sufficient. An early review concluded that, to optimize body composition, dieting athletes should consume 1.8–2.7 g/kg.[18] Later studies have argued that, to minimize lean-mass loss, dieting athletes should consume 2.3–3.1 g/kg (closer to the higher end of the range as leanness and caloric deficit increase).[19] This latter recommendation has been upheld by the ISSN and by a review article on bodybuilding contest preparation.[20,21]

There may also be a reason to eat a little more protein when bulking up. Although gains in lean mass and strength are unlikely to benefit from more than 1.8–2.6 g/kg, a few studies suggest you'll gain less fat if you consume 3.3 g/kg when eating a mildly hypercaloric diet (370–800 kcal above maintenance) and providing a progressive resistance-overload stimulus.[22,23]

It should be noted, however, that people who are overweight or obese and looking to lose weight don't need as much protein as their leaner peers, regardless of activity level. Several meta-analyses involving people with overweightness or obesity suggest that 1.2–1.5 g/kg is an appropriate daily protein intake range to maximize fat loss.[24,25,26] It is important to realize that this range is based on actual body weight, not on lean mass or ideal body weight.

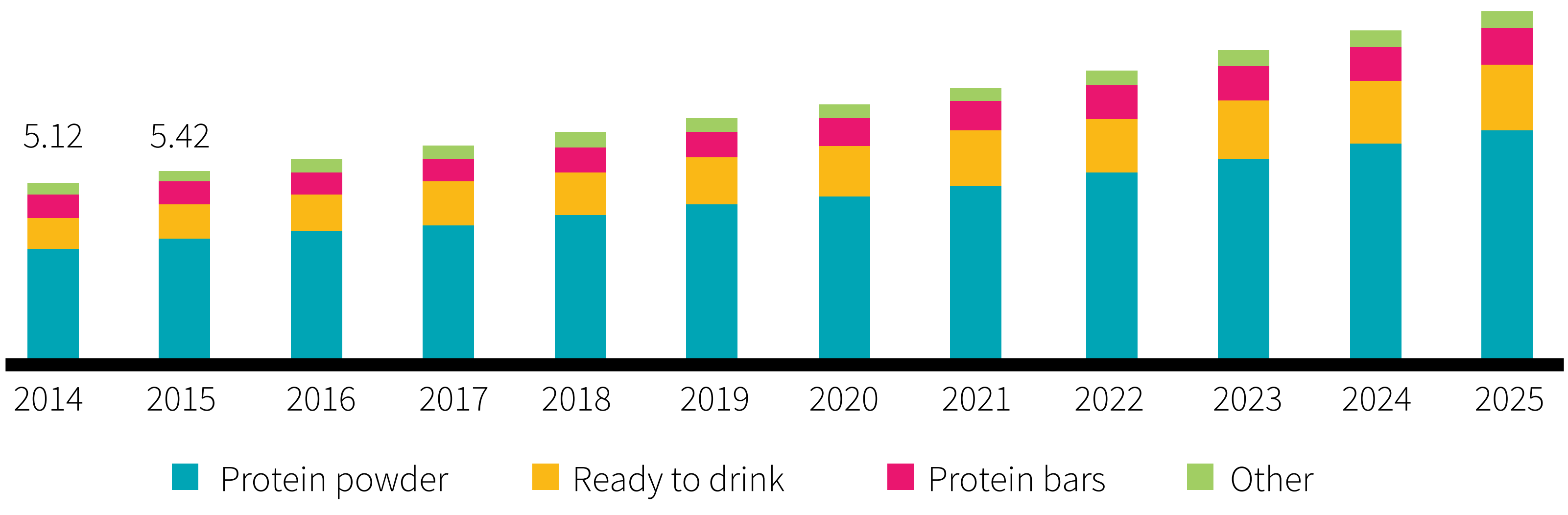
Finally, IAAO studies have suggested that the RDA for pregnant women should be about 1.66 g/kg during early gestation (weeks 11–20) and 1.77 g/kg during late gestation (weeks 32–38).[27,28] When you finally give birth and start lactating, protein requirements are at least 1.5 g/kg.[29,30]

The protein RDA, 0.8 g/kg, represents the *minimal* amount a healthy, sedentary adult needs daily to prevent muscle wasting. Not only does this number not represent an *optimal* intake, but it is based on older assessment techniques. Newer data suggest that 1.2 g/kg is required to avoid deregulation of protein turnover and immune function. And of course, your requirements can increase based on your specific situation (genetics, age, level of physical activity …).

Do you need a protein powder?

Not counting vitamins and minerals, Americans spend over $12 billion a year on supplements, including over $5 billion in protein supplements — mostly powders.

Figure 1: US protein supplements market size, by product (USD billion)



Adapted from https://www.grandviewresearch.com/industry-analysis/protein-supplements-market (accessed June 6, 2019)

The global protein supplement market has a current value of $14 billion, and it is expected to reach $21.5 billion by 2025.[31]

Clearly, many people feel they can't get enough protein from whole foods. But how do protein powders compare with whole foods? What are their advantages and disadvantages? Those are the first questions we need to answer.

More and more people rely on supplements, notably powders, to get their daily quota of protein.

How do protein powders compare with whole foods?

Relevant research is scarce but suggests that the same protein will have essentially the same effect whether it comes from whole food or a protein powder. Pragmatically, you can swap half of your daily whole-food protein for whey protein with no effect on your physical fitness or body composition.[32]

One study pitted casein alone against casein dissolved in milk serum.[33] Another study pitted casein alone against casein taken with milk fat.[34] Neither study found a significant difference in *muscle protein synthesis* (MPS).

And yet, with regard to MPS, studies found whole milk superior to skim milk and whole eggs superior to egg whites.[35,36]

Nutrients other than protein may influence MPS, of course. For instance, depending on how it is processed, whole milk may contain a compound called *milk fat globule membrane* (MFGM). Several studies have reported that supplementation with an isolated MFGM supplement improves physical function in various ways.[37,38,39,40,41]

🔍 **Digging deeper:** How does MFGM affect muscles and nerves?

Unlike other fats, be they animal based or plant based, milk fat is stored in globules. More precisely, it is enclosed in the *milk fat globule membrane* (MFGM), a three-layered membrane composed of proteins, lipids, and numerous minor bioactive components.[42]

One gram of MFGM (as found in about 600 milliliters, or 2.5 cups of whole milk) increases muscle strength, neuromuscular efficiency, and physical function (as assessed through a variety of tests, such as sit-stand and walking distance). This increase is greater when MFGM is combined with exercise.[37–41]

The mechanism behind these benefits is still uncertain, but it may involve neurotransmission. In mice, MFGM increases the expression of *docking protein 7* (DOK7),[43] which is essential to the formation of neuromuscular synapses.[44] In mouse models of neuromuscular diseases, therapeutic upregulation of DOK7 increased muscle strength and motor-unit activity.[45]

It is also possible that the phospholipids present in MFGM are incorporated into the cell membranes of nerves. **In rats**, dietary phospholipids were shown to contribute to the development of the nervous system;[46] **in premature infants**, phospholipid-fortified milk was associated with increased neurobehavioral development;[47] and in **full-term infants**, MFGM-fortified infant formulas led to better cognitive development.[48]

MPS shouldn't be your only concern, of course. Protein's amino acids, separately or combined as biologically active peptides,[49] play vital roles throughout your body; and of course, whole foods contain vitamins, minerals, and other beneficial compounds. There is certainly nothing wrong with incorporating protein powder in your diet, but it should not be your *whole* diet!

You need whole foods for the vitamins and minerals they contain, but you can swap half of your daily whole-food protein for whey protein with no effect on your physical fitness or body composition.

What are the advantages of protein powders?

Getting all your protein from whole foods may be ideal, but it isn't always practical, for at least five reasons: cost, convenience, calories, bioavailability, and appetite.

Cost. Protein for protein, a good protein powder is usually cheaper than whole foods.

Convenience. Cooking takes time. Eating whole foods takes time. And you probably can't do either in your office or at the gym. A protein powder is a quick, non-messy, portable solution.

Calories. In whole foods, protein comes with carbs and fat, so that you may reach your optimal caloric intake before you reach your optimal protein intake.

Table 2: Caloric content of whole-milk powder and whey protein concentrate

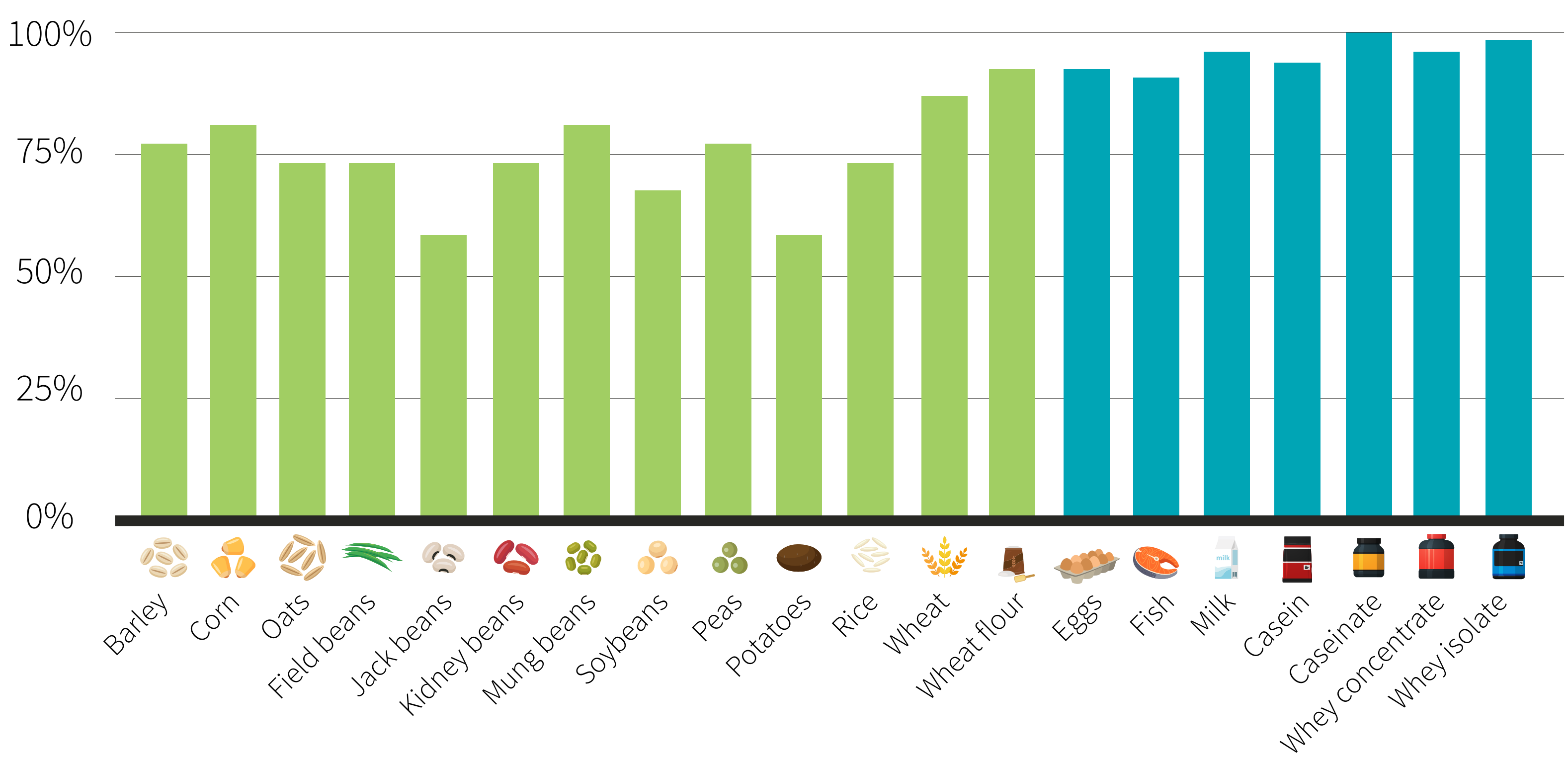
Powder (100 g)	Energy	Protein	Carbs	Fat
Whole-milk powder	496 kcal	26 g	38 g	27 g
Whey protein concentrate	352 kcal	78 g	6 g	2 g

Reference: USDA Food Composition Databases: ID# 01212 and 14066 (accessed May 31, 2019)

Bioavailability. Protein powders bypass several issues of whole-food digestion and absorption that affect protein bioavailability.

This is seldom an issue with animal-based foods, whose proteins consistently demonstrate a digestibility rate greater than 90%, but legumes and grains, the best whole-food plant-based sources of protein, have protein digestibility rates of only 60–80%.[50] In short, your body is better able to use the protein from powders, including plant-based powders, than from plant-based whole foods.

Figure 2: Protein digestibility of various plant- and animal-based proteins



Reference: FAO. Dietary protein quality evaluation in human nutrition. 2013. ISBN:978-92-5-107417-6

In addition, plants contain antinutrients (such as tannins, phytates, and trypsin inhibitors) that inhibit protein digestion and absorption.[51] Cooking only reduces antinutrient concentrations. Plant-based protein powders, however, are mostly free of antinutrients.

Finally, there is one factor that affects the bioavailability of both plant-based and animal-based whole foods: chewing. As a rule, the more we process food by cooking or chewing it, the more digestible it becomes and the more nutrients we can extract.[52]

Whole nuts are a great example of this. Their fat is contained within fibrous cell walls,[53] and the more you chew, the more you can break up those walls,[54] freeing the fat for digestion.

Beef is another example. Because elderly adults wearing dentures have a harder time chewing, they absorb more protein and experience greater whole-body protein synthesis from mince beef than from steak (though the difference is small and MPS rates are similar).[55,56]

Overall, chewing is important to extract calories and nutrients from whole foods. Using protein powders can bypass this requirement entirely. Whether that's a benefit or not depends on your circumstances.

Appetite. Protein is more filling than carbs or fat;[57] some people have trouble hitting their quota because they get too full. These people will find it easier to chug a shake than eat a steak.

Digging deeper: Why is protein satiating?

We know that protein is the most satiating macronutrient, but what gives it this magical power? Well, it turns out that protein acts primarily through appetite-regulation centers in your brain.[58,59] There are many hormones involved (just check out the alphabet soup below), but they can be grouped into **direct** appetite effects and **indirect** appetite effects.

- Protein affects the brain **directly**. Protein is composed of amino acids. In the hypothalamus, amino acids **reduce** the signaling of *neuropeptide Y* (NPY) and *agouti-related protein* (AgRP) and **increase** the signaling of *proopiomelanocortin* (POMC) and *α-melanocyte-stimulating hormone* (α-MSH).

  In this way, amino acids **reduce** the signaling of hypothalamic *AMP-activated protein kinase* (AMPK) and **increase** the activation of the *mammalian target of rapamycin* (mTOR). Since both AMPK and mTOR cooperate as "fuel sensors" in the central nervous system, the end result is increased satiety.[60,61]

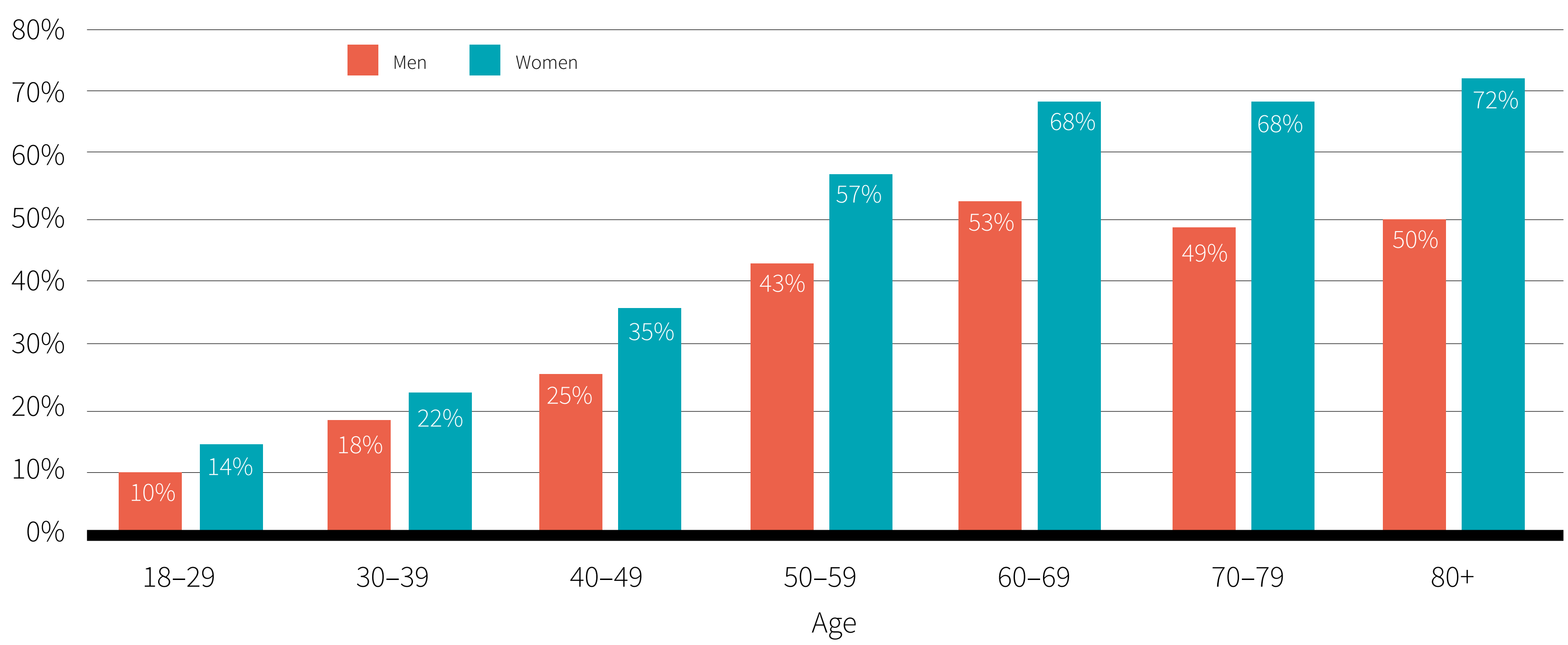
- Protein affects the brain **indirectly** by increasing the rate of glucose production in your gut[62,63] and affecting various intestinal hormones,[64,65] such as *glucagon-like peptide-1* (GLP-1), *peptide YY* (PYY), *cholecystokinin* (CCK), and *ghrelin*. The contribution of intestinal hormones is minor at best, however, since they do not seem to be good predictors of the effect more or less protein will have on total food intake.[66,67]

Finally, for reasons unknown, protein's thermic effect may increase satiety when protein makes up 25–81% of a meal's calories.[57] (Some of the calories in the food you ingest will be used to digest, absorb, and metabolize the rest of the food, and some will be burned off as heat. This process is known under various names, notably *thermic effect of food*).

Low appetite is a common problem for elderly folks, in whom changes in appetite-regulating hormones and neurotransmitters can cause *anorexia of aging*.[68] Loss of appetite is a primary risk factor for developing sarcopenia.[69,70]

Sarcopenia is the age-related loss of muscle mass. In the US, it affects more than 40% of men and 55% of women over the age of 50.[71] It may be the primary cause of physical frailty,[72] which is associated with a higher risk of fractures,[73] falls,[74] hospitalizations,[75] disabilities that affect daily activities,[76] and having to go to a nursing home.[77]

Figure 3: Prevalence of sarcopenia in the US



Reference: Janssen et al. J Am Geriatr Soc. 2002. PMID:12028177

Older men and women require at least 1.2 grams of protein per kilogram of body weight (g/kg) each day to maintain muscle mass,[6,7] while those with sarcopenia need upward of 1.5 g/kg to rebuild lost muscle.[78] Doubling protein intake from 0.8 to 1.6 g/kg was shown to increase lean body mass in elderly men.[79] Similarly, whey protein supplementation was shown to increase lean body mass in elderly women.[80]

Although it is possible to obtain enough protein from whole foods, protein powders have several advantages: they're very low in carbs and fat, cheap (protein for protein), easy to carry, fast to prepare, easy to drink even when you're too full to eat, and easy to digest and absorb. They can especially benefit older people (who often have low appetite and problems to chew) and people who get most of their protein from plant-based whole foods.

What are the disadvantages of protein powders?

The two biggest issues with protein powders are circumstantial and relate to product quality: **tricks** and **contamination**.

Sad to say, but even well-known companies will often try to trick you, usually with a **proprietary blend**. When a company uses a proprietary blend, it doesn't have to disclose the individual amount of each ingredient in the blend. Let's consider two examples:

- Protein blend (Whey protein concentrate, whey protein isolate, whey peptides).

  When you see such a blend, you might picture the ratio as something like 60:30:10. But it can just as easily be 97:2:1, in which case what you get is just expensive whey protein concentrate.
- Protein blend (Whey protein isolate, whey protein concentrate).

  Now that looks better, doesn't it? Since ingredients must be listed in order of weight, you know your protein is more than 50% whey protein isolate. The problem is, the manufacturer may cut costs by using a low-quality whey protein concentrate. Isolates must be at least 90% protein, but concentrates can be anywhere between 29% and 89% protein. So if the proprietary blend is 60% isolate (90% protein) and 40% concentrate (30% protein), the resulting powder is only 66% protein — less than the 80% protein of a decent concentrate.

Of course, even a "pure whey protein concentrate!" product can be a low-quality concentrate.

To avoid falling into either the "proprietary protein blend" trap or the "low-quality whey protein concentrate" trap, look at the food label. Your isolate should be close to 90% protein and your concentrate close to 80% protein. A little lower is all right if the powder is flavored (any flavoring will use a percentage of the powder), but any big discrepancy should stir you away.

Look twice at the serving size! In the US, companies don't have to tell you how much protein you get per 100 grams — only how much you get per "serving". So two products can boast 24 grams of protein per serving even though one has 80% protein (serving size: 30 g) and the other only 69% (serving size: 35 g).

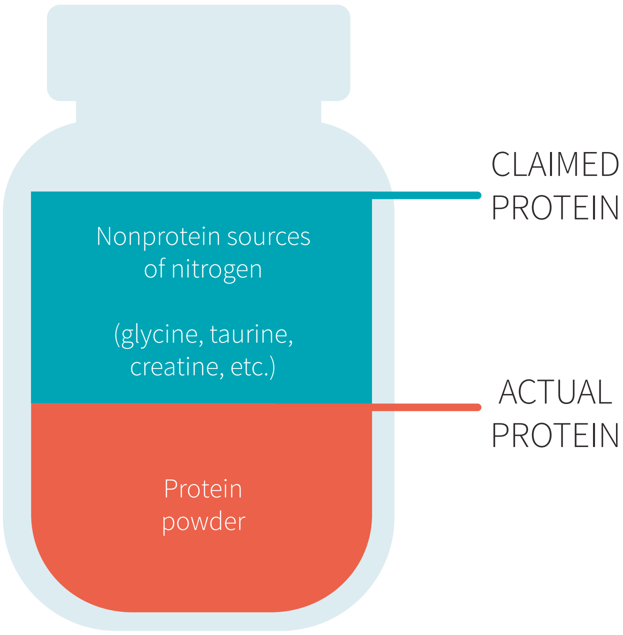
And don't forget to check again. It is not uncommon for a protein powder to launch as a quality product only to be replaced by an inferior version, with no warning or obvious change in packaging, after people have stopped paying attention. One month you may buy a powder with 80% protein, the next you may go buy the "same" powder and discover it has only 69% protein; the company will have changed either the ratios of their proprietary protein blend or the quality of the whey protein concentrate. As we've just seen, they can even keep advertising "24 grams of protein per serving" (in big on the front of the label) just by changing the serving size (printed small on the back).

Okay, that *is* a lot of potential traps. But then, when it comes down to it, if you just look *closely* at protein content and serving size on the food label and do the math, you won't be tricked, right?

Right … unless the manufacturer resorts to **protein spiking.**[81] This trick takes advantage of the way the FDA determines the protein content of powders, which is through a test that measures the powders' nitrogen content. This test works well in theory, because protein *should* be the only ingredient supplying nitrogen.

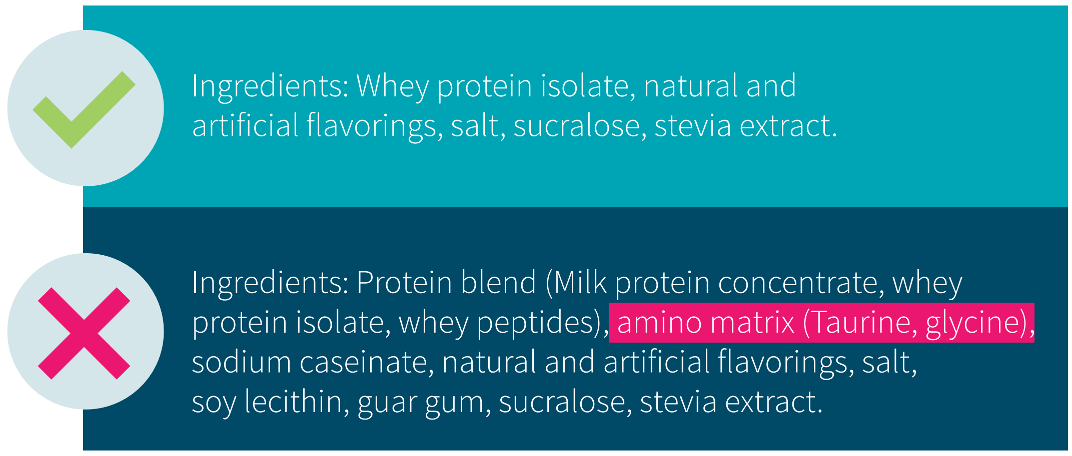
Unfortunately, some manufacturers fill their powders with cheap nitrogen-containing fillers to game the test. These fillers can be any compound that contains nitrogen, such as individual amino acids (glycine, glutamine, etc.) or creatine. In fact, with this testing method, creatine will register as having nearly twice the protein content of whey protein, despite containing no actual protein.

Figure 4: Protein spiking



This trick can be either legal or illegal. **Legally,** companies can include in the list of ingredients a proprietary blend of amino acids. Because the formula is proprietary, they don't have to disclose the individual amount of each amino acid. **Illegally,** companies can simply replace protein with amino acids and not disclose it on the label.

Figure 5: Protein spiking: what it looks like on your label



Aside from purchasing from a reputable supplier, a good rule of thumb is to avoid protein powders that use proprietary blends, especially proprietary blends of amino acids. Additionally, make sure to check the label for the full amino acid profile of the protein powder, if available. In the case of whey protein, the amino acid concentrations should be similar to those in the table below. Small variations are to be expected due to differences in processing methods, but if your whey protein has values that differ greatly from those in this table, something is amiss.

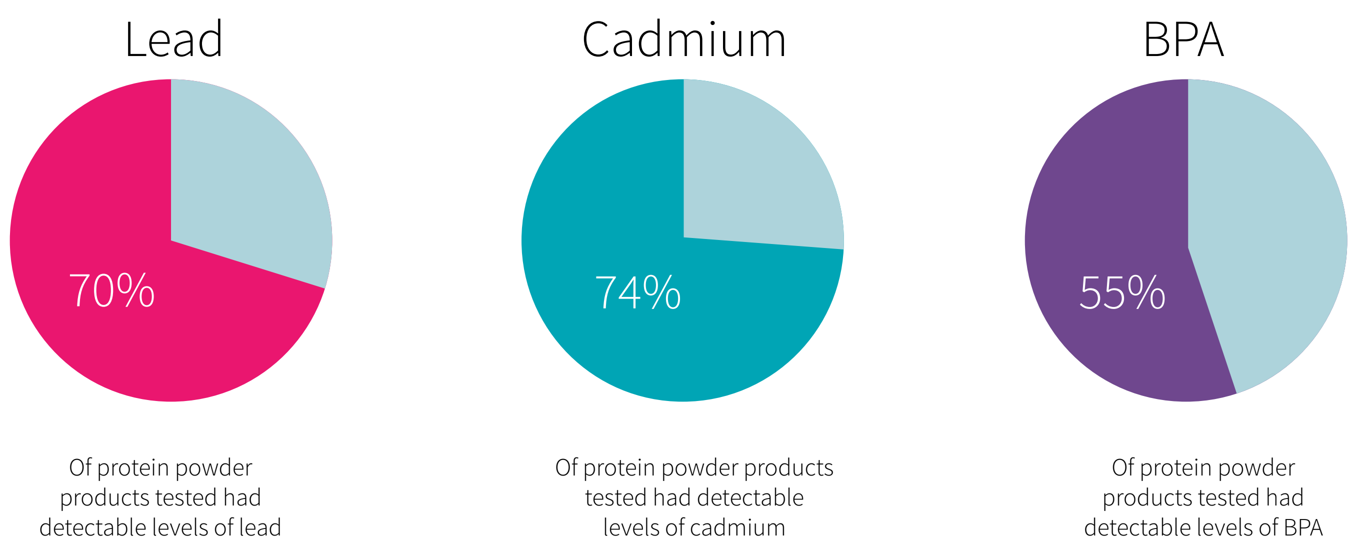
Table 3: Typical *essential amino acid* (EAA) profile of whey protein isolate

EAA	Milligrams per 25 g of protein	% per 100 g of protein
Leucine	2,560	10.3
Isoleucine	1,400	5.6
Valine	1,470	5.9
Histidine	330	1.3
Lysine	2,425	9.7
Methionine	420	1.7
Phenylalanine	645	2.6
Threonine	1,980	7.9
Tryptophan	470	1.9

Reference: Kalman DS. *Foods.* 2014 Jun.

Contamination is another issue that can affect protein supplements,[82] even those sold by well-intentioned companies. For example, third-party testing by the Clean Label Project found that, among 134 tested protein powders, 70% had detectable levels of lead, 74% had detectable levels of cadmium, and 55% had detectable levels of *bisphenol A* (BPA).

Figure 6: Clean Label Project protein powder study results, 2018



Adapted from https://www.cleanlabelproject.org/protein-powder (accessed June 6, 2019)

Contaminants, such as heavy metals and plastic derivatives, can make their way into protein powders by way of ingredient sourcing and manufacturing practices. Since supplement companies are not required to test their products for contaminants, they are left to voluntarily do so in order to boost transparency, consumer trust, and perception of quality. But testing is expensive, and the return on investment may be poor.

Several third-party companies test dietary supplements for quality, purity, potency, and composition, so doing a quick search to see if a company's products have been tested may be worthwhile. Doing a little background research before purchasing a product is generally a good idea.

Tip: Safe supplements

First, if a protein powder has caught your interest, visit the manufacturer's website. Does the manufacturer use specific manufacturing protocols? What are its in-house quality-control practices? Is it transparent with its practices and findings? Does it test for contaminants?

Second, check if the manufacturer has received warning letters from the FDA.

Finally, check if the product you're interested in, or other products from the same manufacturer, has been tested by third parties such as ConsumerLab (products that pass its tests may display the CL Seal), NSF International (products that pass its tests may display the NSF mark), the United States Pharmacopeia (products that pass its tests may display the USP Verified Mark), or the International Society for Pharmaceutical Engineering (which checks for compliance to Good Manufacturing Practice, or GMP).

And of course, don't forget to check the product's label for known potential allergens.

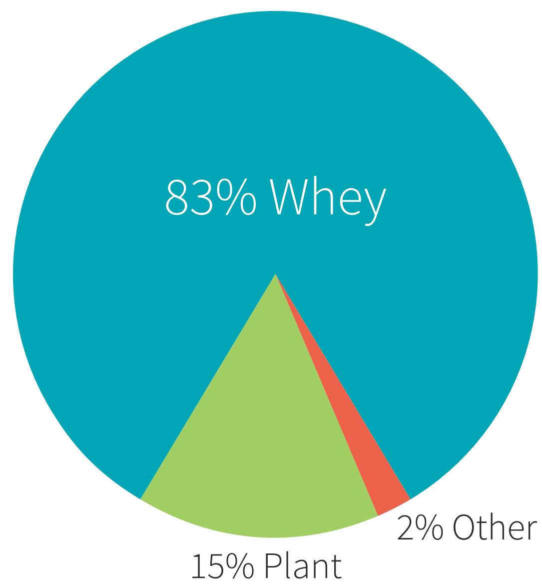
Protein powders have three main *potential* disadvantages: proprietary blends designed to trick you, protein spiking, and contaminants (such as heavy metals). Avoid proprietary blends and research the quality-control practices of a company before purchasing its products.

Whey protein

And now for our good friend whey.

Depending on the stage of breast milk production, the protein in human milk is 80% to 50% whey.[83] By contrast, the protein in cow's milk is 20% whey (and 80% casein). Whey protein from cow's milk makes for more than 80% of the protein powder market.

Figure 7: Online sales of protein powders



Reference: 1010data market insights report: online protein powder category (Nov. 2015 – Oct. 2016). 2017

Whey protein is the most popular type of supplemental protein and the gold standard in sports nutrition research — for good reason. Several rating systems have been developed over the years to rank proteins based on two criteria: bioavailability and *essential amino acid* (EAA) composition. No matter the method, whey beats out most other protein sources.

Table 4: Comparison of various foods' protein quality

	PER	BV	NPU	PDCAAS
Whey	3.2	104	92	1.00
Casein	2.5	77	76	1.00
Milk	2.5	77	76	1.00
Egg	3.9	100	94	1.00
Beef	2.9	80	73	0.92
Soy	2.2	74	61	1.00
Black beans	0	-	-	0.75
Wheat gluten	0.8	64	67	0.25

PER: Protein Efficiency Ratio | BV: Biological Value | NPU: Net Protein Utilization |
PDCAAS: Protein Digestibility-Corrected Amino Acid Score

Whey is high in EAAs, notably leucine (the most anabolic amino acid).[84] Whey protein is 52% EAAs and 13.6% leucine. By contrast, protein from other animal sources is roughly 40–45% EAAs and 7–8% leucine, while protein from plant sources is even lower.

Further, whey protein is rapidly digested and absorbed, and so is an ideal companion to resistance training, since rapid increases in serum EAAs lead to greater MPS compared with slower, more steady rises.[85,86] (Unsurprisingly, consuming whey protein away from training sessions lessens its benefits.[87])

Figure 8: EAA content of plant- and animal-based proteins

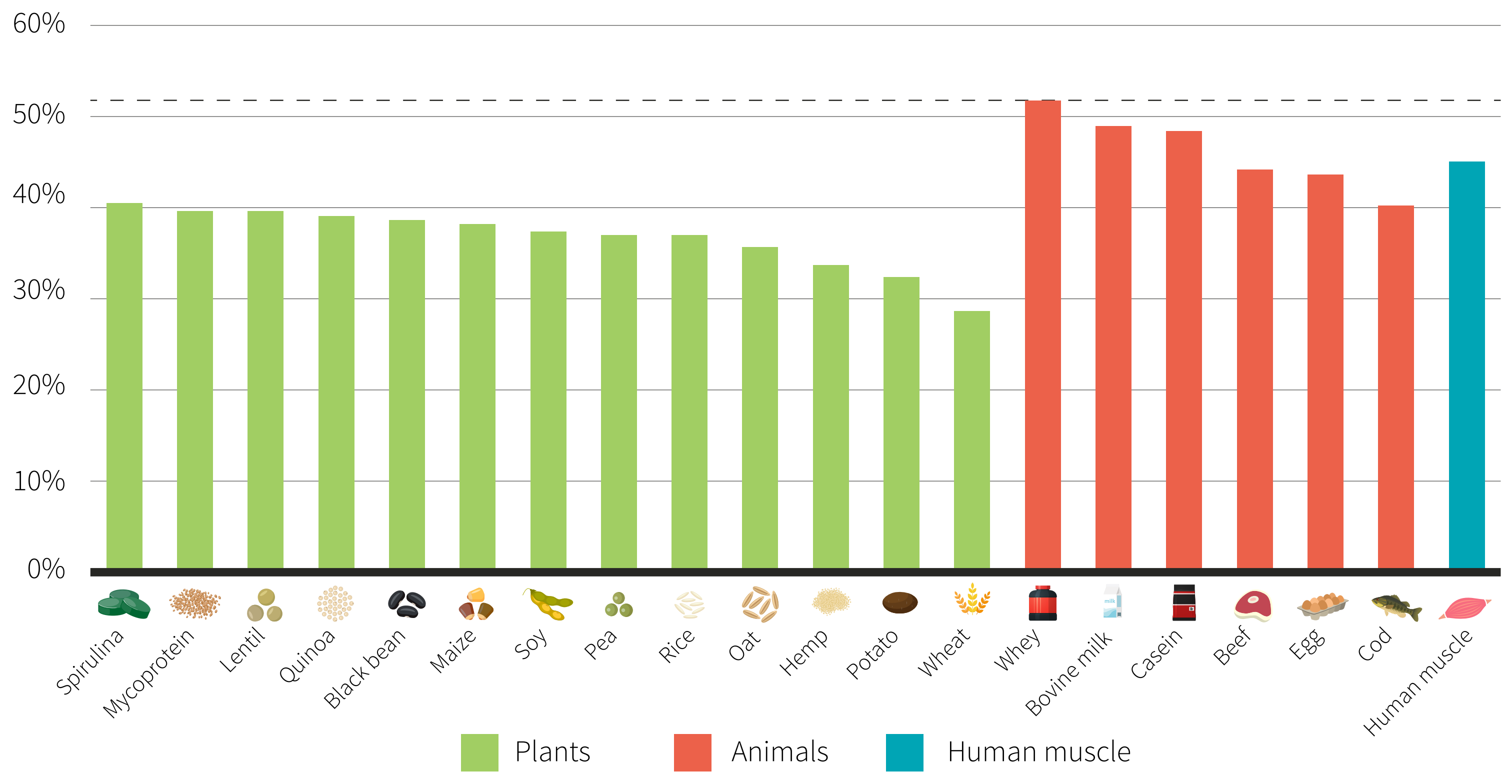
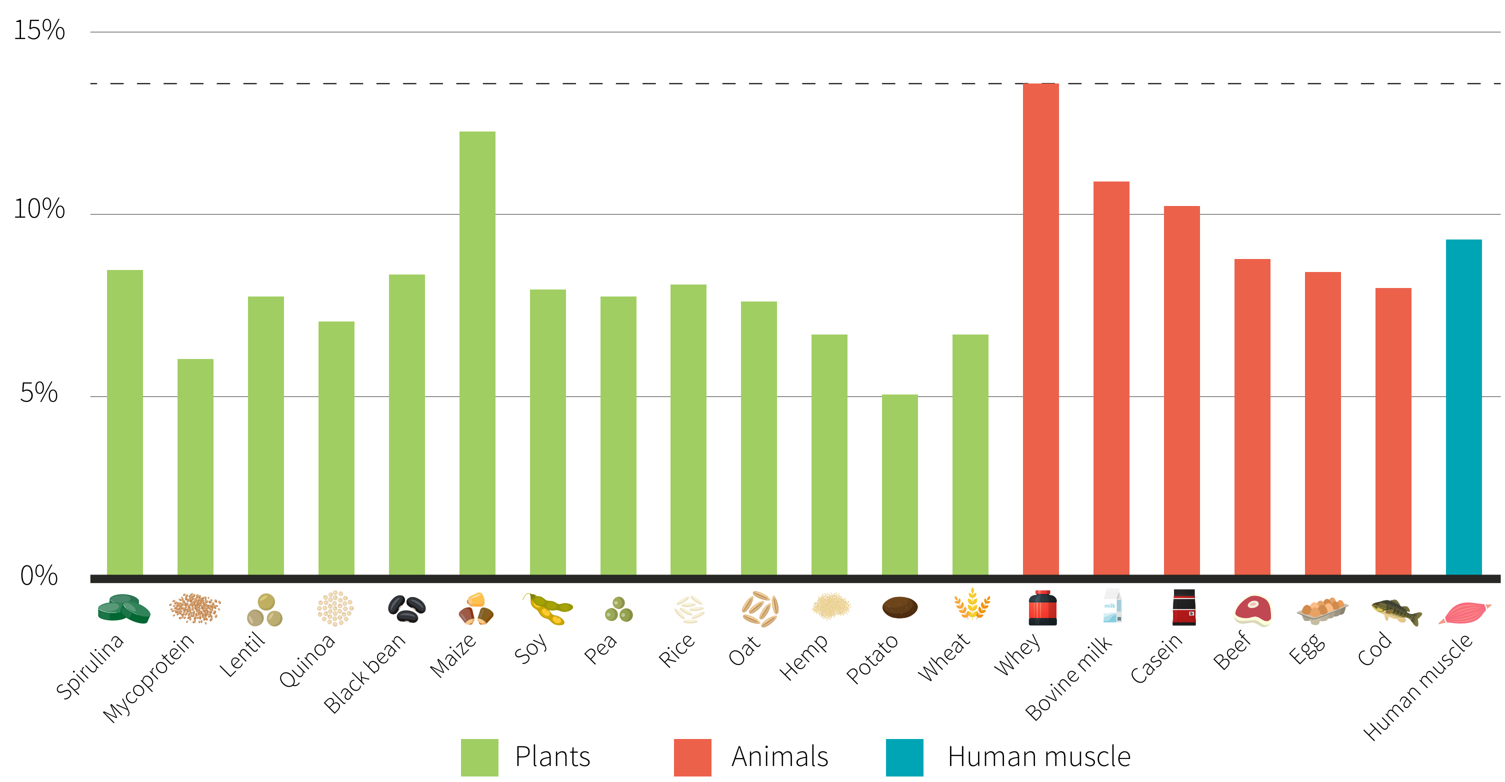


Figure 9: Leucine content of plant- and animal-based proteins



Whey protein is the gold-standard protein in sports nutrition. It has a great amino acid profile, is highly bioavailable, and is an ideal companion to resistance training due to it quick digestion and absorption.

Bioactive peptides

Proteins are a source of amino acids, but also of peptides — small chains of amino acids that aren't completely broken down by your digestive enzymes. Some of those peptides, called *bioactive peptides*, have physiological effects.[49]

A peptide can become bioactive only after being freed from a larger protein strand (a protein subfragment) by your digestive enzymes. It can act locally in the gut or be absorbed into the bloodstream,[88,89,90] from where it has a variety of effects on tissues throughout the body.

Whey protein is a rich source of bioactive peptides that benefit the cardiovascular and immune systems.[49,91] These peptides may partly explain why breastfed infants have a lower risk of developing obesity, diabetes, and cardiovascular disease than formula-fed infants:[92] infant formulas are not always made with dairy (soy is a popular alternative), but even when they are, cow's milk has less whey than human milk, and what whey it does have may have been denatured during processing.

Table 5: Bioactive peptides from whey protein

Whey subfraction	% of whey protein	Potential health effects
β-lactoglobulin	56–60%	Might lower blood pressure and strengthen the immune system.
α-lactalbumin	18–24%	Might benefit cognition and reduce blood pressure. Also possesses immune-modulating, antimicrobial, antiviral, and antioxidant properties.
Immunoglobulins	6–12%	Might strengthen the immune system.
Serum albumin	6–12%	Rich source of glutathione precusors.
Lactoferrin	1–2%	Might reduce appetite and body fat while benefiting glycemic control. Strengthens the immune system.
Lactoperoxidase	0.5–1%	Has antibacterial properties.

Whey protein contains bioactive peptides that are released during digestion. These peptides can act locally in the gut or be absorbed into the bloodstream. Some appear to benefit the cardiovascular and immune systems.

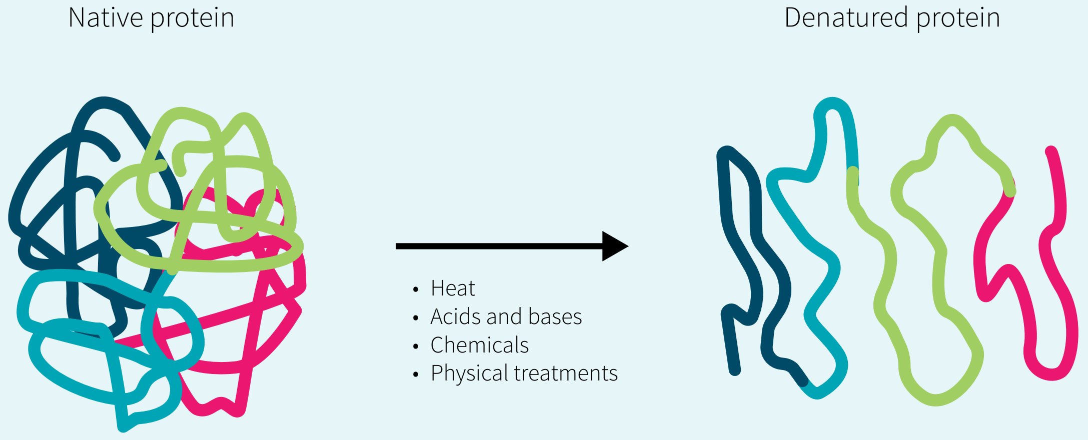
Whey processing and denaturation

The overall process of making whey protein powders is rather simple, yet with each step there can be differences in manufacturing. Some techniques can denature the protein,[93] preventing the formation of bioactive peptides when the protein is digested.

Digging Deeper: Denaturation

Denaturation is the alteration of a native structure. Protein denaturation can be caused by heat, chemicals, acids, or physical treatments. It changes the way the protein functions and interacts with your body.

Figure 10: Protein denaturation



Denaturation is an important part of our evolutionary history. It was our ability to harness heat to denature the physical structures of food that allowed us to obtain more calories from what we ate.[52] For example, cooking an egg denatures its proteins in such a way as to increase their bioavailability from roughly 50% to 90%.[94]

As discussed earlier, however, the digestion of whey protein leads to the production of bioactive peptides that can benefit notably your immune system. The denaturation of whey protein can interfere with the production of those peptides because a change in the protein's structure means that our digestive enzymes will act on it differently.

Denaturation of whey protein may explain, at least in part, why observational studies have found that infants drinking milk that has been aggressively heated are at higher risk of allergic diseases and respiratory infections than infants drinking raw milk or milk that has not been aggressively heated.[95,96] Raw milk can also protect mice against the development of asthma, while heated milk cannot.

If your goal is exclusively to get protein, then this may not matter. But since bioactive peptides may benefit your health, then why not opt for a whey protein that supplies them? It's like icing on the cake.

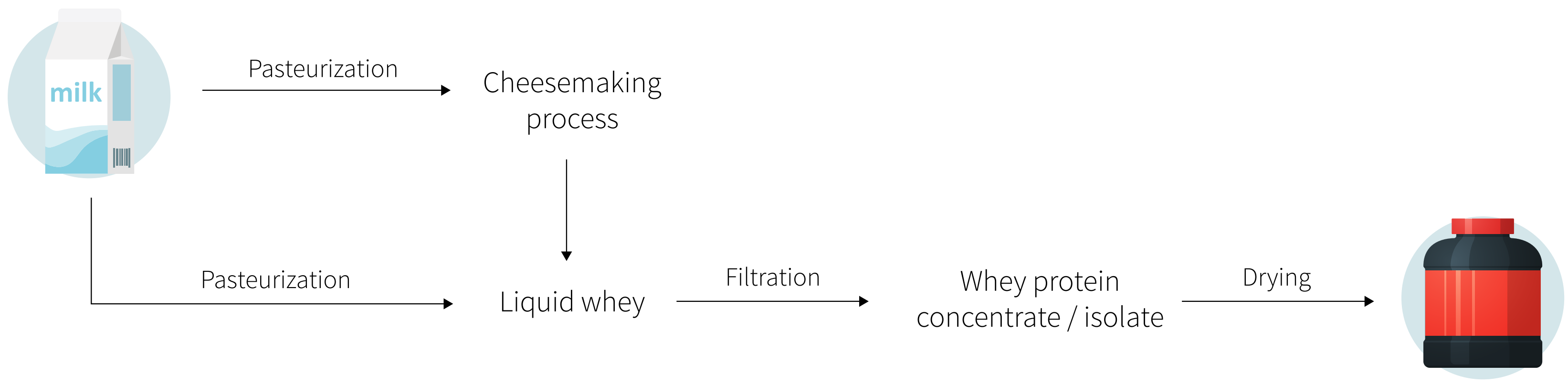
The first step in making a whey protein powder is securing a source of milk. There are many companies now advertising whey protein obtained from pastured cows fed an ecologically appropriate diet, and whether the cattle's diet and life conditions affect whey quality will be discussed later.

From milk, the liquid whey can be extracted. It is most often a by-product of cheesemaking, although an increasing number of companies are extracting their whey directly from milk. The differences between the two types of liquid whey, *cheese whey* and *native whey*, are used as advertising arguments by manufacturers; we'll discuss them later.

Both types of liquid whey need to be filtered to separate the whey protein from the lactose, fat, bacteria, and other unwanted components. There are several methods for accomplishing this, which we'll also discuss.

Finally, the whey extract is dried into a powder and sold as *whey protein concentrate* (29–89% protein) or *whey protein isolate* (at least 90% protein). Further processing with enzymes that "predigest" the protein produce *whey protein hydrolysate*. These three forms of whey protein will also be discussed.

Figure 11: From milk to whey protein powders



Whey protein powders are created through the filtration, concentration, and drying of liquid whey obtained either as a by-product of cheesemaking or directly from milk. At each step in production, differences in manufacturing practices can denature the protein and thus prevent your digestive enzymes from forming bioactive peptides out of it.

Sourcing the whey: does *organic* or *grass-fed* matter?

First, let's deal with that organic certification you might see on a whey protein powder. It means that the cow was given neither hormones nor antibiotics, and that its pasture or feed was itself organic. Does that make your powder healthier? Maybe, maybe not: while there is some evidence that organic produce might be safer, whey protein is very different from a salad. All we can tell is that there doesn't appear to be any difference in the whey protein composition of the milk produced by two farms, one certified organic and the other not, that have similar farming practices.[97]

But what about when the farming practices differ?

An increasing number of companies advertise that they source their whey from cows raised on pasture or fed grass rather than grain. There are important environmental and ethical arguments to be made about either practice, but our focus here will be on its effect on the nutritional value of whey protein.

This effect is, at best, minimal. An early study reported that greater access to pasture resulted in small increases in some whey bioactive peptides, but small decreases in others,[98] whereas a later study found no meaningful differences.[99]

There is little *nutritional* difference between whey protein sourced from the milk of cows raised conventionally and whey protein sourced from the milk of cows pastured or grass fed. Also, an organic certification has no impact on whey protein composition.

Does pasteurization denature whey?

The FDA requires that all milk intended for human consumption be pasteurized, including any used to make whey protein powders. So all whey protein powders are pasteurized at least once, meaning there is no such thing as raw whey protein powder.

The most common type of pasteurization in the dairy industry is *high-temperature, short-time* (HTST) pasteurization, in which milk is heated at 72°C (161°F) for 15 seconds and then cooled rapidly. Basically, milk is run through millimeter-wide, superheated tubes for 15 seconds, then through supercooled tubes to end the pasteurization process nearly instantly.

HTST pasteurization does not denature whey protein,[100,101,102,103,104] which is why it is used notably in the production of a patented, nondenatured whey protein powder.

A less common form of pasteurization, called *vat* or *low-heat* pasteurization, involves heating large batches of milk to 63°C (145°F) and holding them at that temperature for 30 minutes. Some companies may advertise the use of this type of pasteurization because it uses lower temperatures than HTST pasteurization.

Over time, however, this "low" heat is still high enough to denature several whey protein subfractions,[105] especially when we consider that the exposure time is not just the 30-minute holding temperature but also the time it takes to heat and cool the vat of milk. Some studies have reported that 10–20% of whey proteins are denatured during vat pasteurization.[106,107]

In fact, this is the primary reason why many cheesemaking plants turn away batches of vat-pasteurized milk: denatured whey protein sticks to the casein, negatively affecting cheese quality.[108] To quote *Cheese Reporter*,[108]

> Tera Johnson, CEO of the new whey plant being constructed in Reedsburg, WI, said they cannot use the whey from cheese plants where batch pasteurizers are used, as the whey had undergone too much denaturation.

The FDA requires that all milk intended for human consumption be pasteurized, including any used to make whey protein powders. *High-temperature, short-time* pasteurization does not denature the milk's whey protein, unlike the less common method called *vat* or *low-heat* pasteurization.

Cheese whey vs. native whey

When whey is a by-product of cheesemaking, it is called *cheese whey*. When it is extracted directly from milk, it is called *native whey*. Most supplement companies use cheese whey; those that use native whey claim that it is superior because it has more leucine and because the heat and chemical processes used to make cheese can denature the whey protein.

They aren't *technically* wrong.

Native whey does contain marginally more leucine than does cheese whey: 2.7 versus 2.2 grams per 20 grams of protein. But one study comparing the two types of whey protein found similar increases in anabolic signaling, MPS, and strength recovery in resistance-trained young adults,[109] while another found similar rates of anabolic signaling and MPS in elderly adults.[110]

And yes, cheesemaking *can* denature whey protein. Whey can be obtained from different types of cheeses. To produce acidic cheeses (cottage cheese, cream cheese, etc.), the milk is exposed to high temperatures and its acidity is altered chemically. Since both processes can denature whey protein, you should avoid using powders made from *acid whey*.

Thankfully, most cheese whey comes from the production of natural, rennet-produced, cultured cheeses (Cheddar, mozzarella, etc.).[111] Milk is allowed to ripen for a mere 60 minutes after being mixed with lactic acid bacteria,[112] at which point the enzyme rennet is added to the mixture for another 60 minutes before the liquid whey,[113] called *sweet whey*, is drained. Both exposures are too brief — and take place at about half the temperature required — to denature whey protein. A patented non-denatured whey protein powder is made from sweet whey.

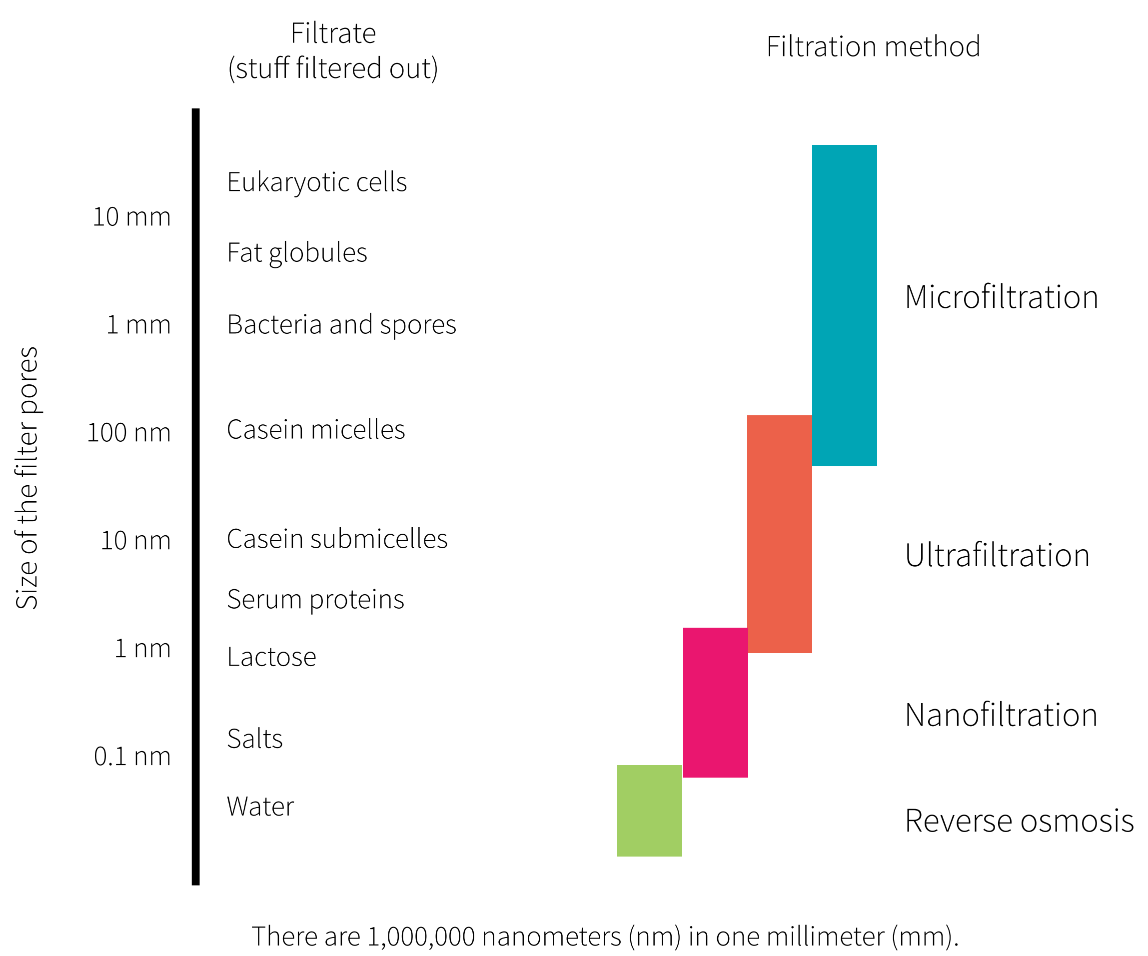
Whey protein can be obtained as a by-product of cheesemaking (cheese whey) or directly from milk (native whey). Native whey is a little richer in leucine, but the difference doesn't seem to have real-life significance. Native whey isn't denatured, but neither is sweet whey, the most common kind of cheese whey. Acid whey, another kind of cheese whey, is denatured and should therefore be avoided.

Does the filtration method matter?

Liquid whey, whether obtained as a by-product of cheesemaking or directly from milk, contains more than just protein. Filtration is required to remove unwanted components such as bacteria, fat, lactose (milk sugar), and residual casein (the other protein in milk). Sweet whey, the most common type of liquid whey, is 5.14% carbs (lactose), 0.85% protein (mostly whey), and 0.36% fat.

The most common method used in making whey protein powders is *ultrafiltration*, sometimes paired with *microfiltration*. Both methods involve pushing liquid whey through a membrane without the use of heat or chemicals;[114] the whey protein passes through nondenatured, while unwanted components do not pass at all.[115]

Figure 12: Four filtration methods



Adapted from Kumar et al. *Asian-Australas J Anim Sci*. 2013. PMID:25049918

Some companies may advertise the use of *crossflow filtration*. With this method, water gets run across the membrane rather than directly onto it, thus removing built-up gunk. This results in increased throughput — which only benefits the manufacturer.

Ion exchange is another method of isolating protein: liquid whey is run through special resins that chemically bind the protein.[116] Alas, the chemicals alter the pH of the liquid whey and denature the whey protein.[117,118] *Mixed-matrix-membrane ion exchange*, a novel ion-exchange method developed as part of a PhD thesis back in 2003,[119] does without harsh chemicals and can thus yield nondenatured whey protein,[120] but it has received little attention;[121] it doesn't seem promising enough to be favored over the membrane filtration methods.

Microfiltration and ultrafiltration do not denature whey protein, whereas ion exchange does.

Is spray drying a concern?

Spray drying is the method most widely used in the dairy industry to convert the whey protein solution into a powder. This process can take on many forms, some of which denature whey protein.

Conventional methods, which maximize production speed, denature 30–40% of the protein, even at what would be considered low temperatures for spray drying: 60–80°C (140–176°F).[122] At those low temperatures, lower-throughput spray drying does not denature whey protein.[123]

To avoid the issue of heat denaturation entirely, some manufacturers use vacuum drying (below room temperature) and/or freeze drying.[124] Such processes, which are used notably in the making of a patented nondenatured whey protein, take much longer (15–18 hours, usually) and so are not a prime choice for the mass production of whey protein powders.

Freeze drying and vacuum drying do not denature whey protein. Spray drying can, but steps can be taken so it won't.

Whey protein concentrates and isolates

Whey protein powders can be divided into three main categories: concentrates, isolates, and hydrolysates. Here, we focus on concentrates and isolates. Hydrolysates are discussed next.

Whey protein concentrates and isolates differ mostly in their protein, lactose, and fat contents. Isolates are at least 90% protein by weight; therefore, they contain with very little lactose and fat. Concentrates, the most widely used form of whey protein in food manufacturing, contain 29–89% of protein by weight. The most common types of concentrate in the US, used notably for food aid,[125] are WPC34 (34% protein) and WPC80 (80% protein). WPC80 is the protein powder most commonly used by the supplement industry.

WPC80 and whey protein isolates are very similar. The former contains a little less protein and a little more lactose and fat, but that's it. Since isolates are significantly more expensive, a decent concentrate will usually be your better choice, unless you are *very* sensitive to lactose.

There is little difference between a whey protein isolate (90% protein) and the most common type of supplemental whey protein concentrate (80% protein). An isolate is seldom worth its higher price tag, unless you are *very* sensitive to lactose.

Whey protein hydrolysates

Whey protein hydrolysates are concentrates or isolates that have been "predigested"—meaning that the protein has been broken down into peptides (hydrolysates), primarily through enzymatic means.[126]

Hydrolyzation denatures whey protein, but whether this denaturation results in higher or lower bioavailability of the bioactive peptides depends on the type and amount of enzymes used, as well as on the incubation temperature, pH, and time.[127]

From a practical standpoint, hydrolyzed whey protein is often promoted as being better for building muscle because it is absorbed faster than other forms of whey protein. But that isn't so. In several studies, hydrolysates increased serum concentrations of amino acids faster than did concentrates,[128] but to a similar or even lower extent than did isolates.[129,130,131] More to the point, two separate studies reported that concentrates and hydrolysates, coupled with resistance training, led to similar increases in strength and muscle mass.[132]

Whey protein concentrates and hydrolysates have similar effects on muscle mass and strength, but hydrolysis denatures the proteins.

What if I'm allergic to whey?

First, make sure it isn't the lactose you're sensitive to by trying a whey protein isolate.

If you do find that you are sensitive to (or even allergic to) whey protein, then the cause is probably β-lactoglobulin,[133] the main whey protein subfraction in cow's milk. That's because this protein is absent from human milk. Other whey protein subfractions can be responsible for the allergy or sensitivity, but it is less likely.

Milk from buffalo, sheep, goats, horses, and donkeys also contains β-lactoglobulin,[134] and cross-reactivity between species is common (so if you are sensitive to cow's milk, don't be surprised if you are also sensitive to goat's milk).[133] Camel milk lacks β-lactoglobulin, but finding a whey protein powder sourced from camel milk may prove … challenging.

If you think you're allergic to whey protein, first rule out lactose by trying an isolate. If you still have a reaction, the problem may instead be β-lactoglobulin; try to find some camel's milk, which doesn't contain this protein subfraction.

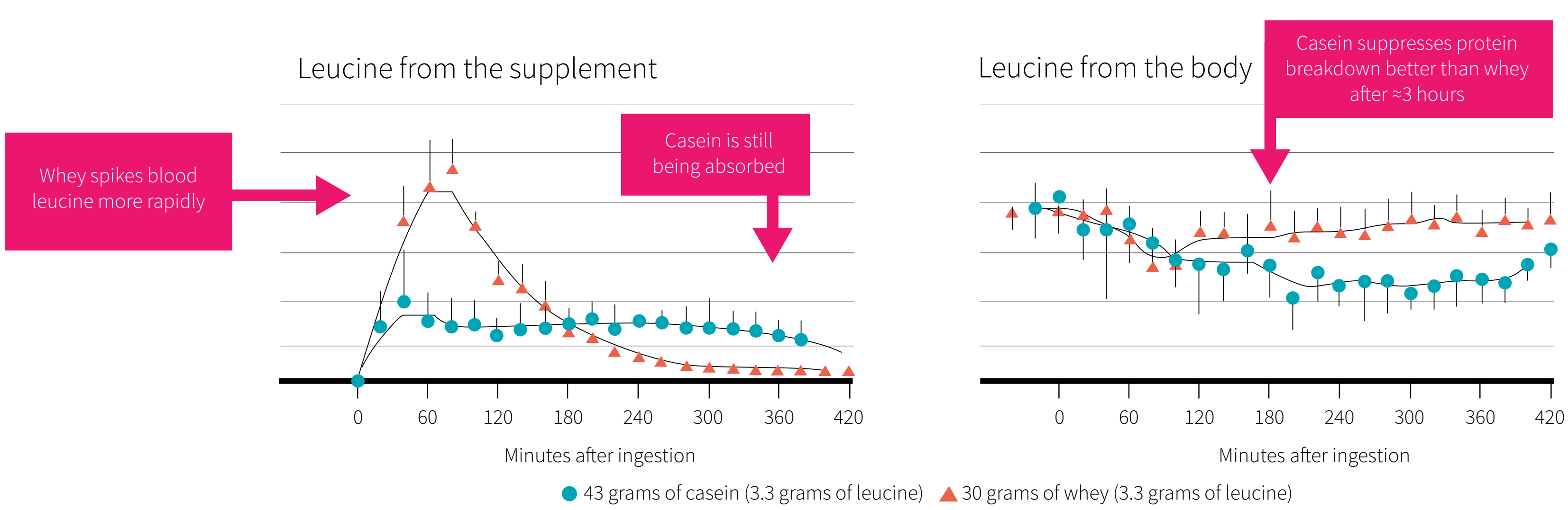
Casein

Casein accounts for 80% of the protein in cow's milk (compared to 30% in human milk, 23% in goat's milk, and 15% in the milk of sheep and buffalo). It is essentially the only protein in cheese (coagulated casein plus milk fat) and strained yogurt (a.k.a. Greek yogurt).

Compared to whey protein, casein is lower in *essential amino acids* (EAAs), notably leucine, and so has lower biological quality. Its speed of digestion and absorption may also be lower, depending on the type of casein powder you choose: micellar casein, casein hydrolysates, and caseinates.

Micellar casein is the form of casein found in milk. It digests very slowly: consuming 40 grams can maintain elevated levels of serum EAAs, notably leucine, for 6–7 hours (compared to about 4 hours for whey).[135,136] This is because, under acidic conditions (as found notably in your stomach), micellar casein coagulates into a blob that is difficult for your digestive enzymes to break down.[137] Unfortunately, slower digestion speed means less stimulation of MPS with micellar casein than with whey protein.[138,139]

Figure 13: Leucine appearance in the blood



Reference: Boirie et al. *Proc Natl Acad Sci USA*. 1997 Dec

The two other forms of casein, caseinates and hydrolysates, are created by destroying the micellar structure of casein, allowing for faster digestion.[137,140] Their digestion speed is, in fact, very similar to that of whey protein. However, whey protein, being richer in EAAs and leucine, still stimulates MPS more than do caseinates and hydrolysates during the first 3 hours after ingestion (and similarly thereafter).[141,142]

Casein has less EAAs, notably leucine, than does whey protein. Micellar casein digests slowly, whereas caseinates and casein hydrolysates digest quickly. None of these forms stimulate MPS more than does whey protein.

Bioactive peptides

Like whey protein, casein is composed of subfractions that form bioactive peptides when digested.[143] In casein, those subfractions are α-, β-, and κ-caseins, from which your digestive enzymes can form peptides that stimulate opioid pathways and benefit your immune and cardiovascular systems.[49,144] Of those peptides, *glycomacropeptide* (GMP) and the β-casomorphins (BCMs) are of special interest.

GMP exists naturally in small amounts in casein powder but comes about primarily through the digestion of κ-caseins. It acts as an antimicrobial, strengthens the immune system, and benefits dental health.[145]

BCMs, which are produced during the digestion of β-caseins, are the peptides with opioid, or morphine-like, properties. Of the various BCMs, only BCM-7 has been heavily investigated, because of associations found with higher risks of certain disorders and diseases, such as autism, cardiovascular disease, and type I diabetes. However, a comprehensive review by the European Food Safety Authority concluded that those associations were based largely on speculation and somewhat conflicting explanations, suggesting that more research into the role of BCM-7 in human health is required.[146]

Several bioactive peptides from casein can benefit your health, but BCM-7, which stimulates your opioid pathways, has been linked to various diseases based on weak scientific evidence. More research is required.

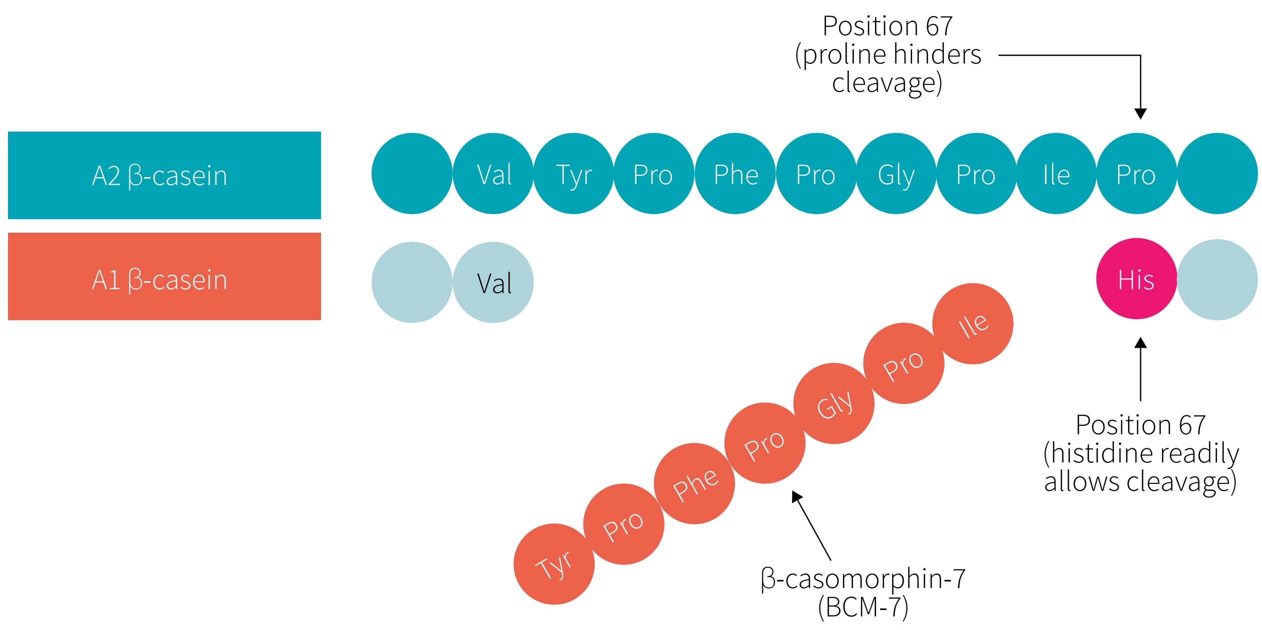
A1 vs. A2 β-casein

There are two types of β-casein protein subfraction: A1 and A2.[147]

A2 is the natural and original form of β-casein. It is the form found in the milk of humans, goats, sheep, and purebred Asian and African cattle. The A1 variant, a genetic mutation, appeared in European cattle about 5,000 years ago. Due to crossbreeding, most dairy products contain both A1 and A2 (both are present in the milk of prominent cattle breeds such as Ayrshire, Guernsey, and Holstein).

The practical difference between the two types of β-casein is that your digestive enzymes can form BCM-7 out of A1, not A2.

Figure 14: Difference between A1 β-casein and A2 β-casein



Although the role of BCM-7 in actual diseases is uncertain, there is consistent evidence from animal studies that consuming A1 promotes inflammation through the binding of BCM-7 to opioid receptors in the gut.[148] Human data are scarce, but the few studies available suggest with moderate certainty that A1 is proinflammatory.[149,150,151,152,153]

In fact, some of these studies suggest that people who believe they are lactose intolerant are actually sensitive to A1 instead; they do not report symptoms of lactose intolerance when drinking milk that contains A2 only.

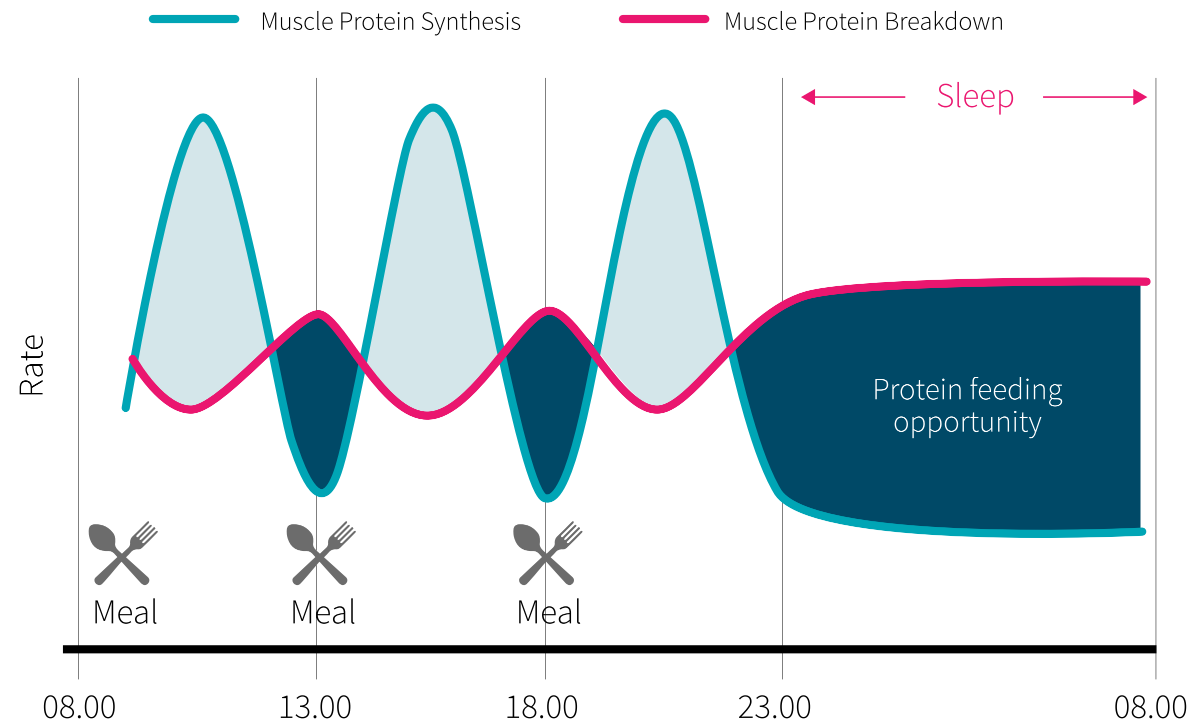
Note that any possible proinflammatory effect may be masked by milk's high nutritional value, since both types of β-casein similarly benefit exercise recovery.[154] If you feel "off" with a regular casein powder, you may want to try one that contains only A2.

A1 β-casein may be more inflammatory than A2 β-casein, but research is preliminary. People who suspect they are sensitive to A1 can try casein powders from animals that produce only A2, such as goats and sheep.

Is there a benefit to taking casein before bed?

The slow digestion of micellar casein has led to the idea that taking this protein before bed could benefit muscle mass and exercise recovery by providing the body with a steady flow of amino acids during a time when fasting normally dominates.[155]

Figure 15: Effects of meals and sleep on muscle protein synthesis and breakdown



Serum levels of EAAs, notably leucine, do stay elevated longer with micellar casein (6–7 hours) than with whey protein (about 4 hours),[136] but those EAAs get incorporated into muscle tissue only for the first 3–4 hours.[136,156] In other words, micellar casein and whey protein increase *muscle protein synthesis* (MPS) for the same length of time.

Three pertinent studies looked at bedtime casein: the first used a casein mix (half micellar, half hydrolyzed); the second, micellar casein; the third, caseinates. Remember that hydrolysates and caseinates digest as fast as whey protein, so much slower than micellar casein.

- Two groups of young men took a powder daily near bedtime — one group took 28 grams of a slow/fast casein mix, the other a placebo. After 12 weeks, the casein group had experienced greater increases in strength and muscle mass.[157]
- Two groups mixing resistance-trained men and women took 54 grams of micellar casein daily — one group in the morning, the other near bedtime. After 8 weeks, strength and body composition still hadn't changed in either group.[158]
- Two groups of resistance-trained young men took 35 grams of caseinates daily — one group in the morning, the other near bedtime. After 10 weeks, strength and muscle size had increased similarly in the two groups.[159]

We notice two things. First, that the two studies that used a fast-digesting protein saw benefits, whereas the study that used solely a slow-digesting protein did not. Second, that the two studies that compared morning and bedtime proteins saw no difference in results.

Although comparing studies with different protocols is always iffy, it seems that speed of digestion matters, whereas time of ingestion does not.

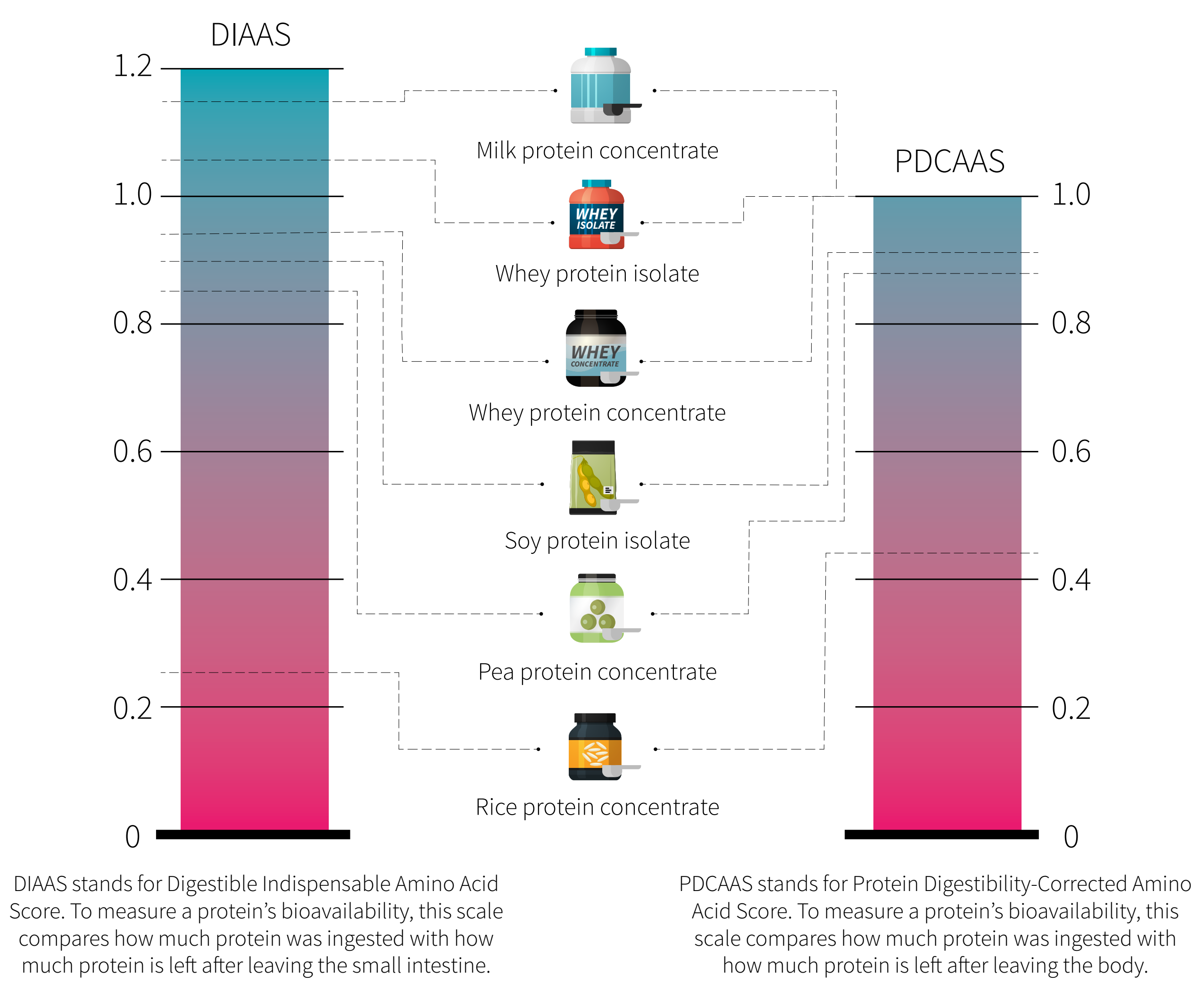
If you haven't consumed enough protein during the day, then taking casein before bed can benefit you, but the same can be said of any other protein.[155] Time of ingestion doesn't seem to matter: whether in the morning or near bedtime, a fast-digesting protein, such as whey protein, seems a better choice for increasing strength and muscle mass than a slow-digesting protein, such as micellar casein.

Milk protein concentrates and isolates

In theory, combining EAA-rich, leucine-rich proteins that have different digestion speeds should offer the best of both worlds: a rapid *and* sustained anabolic response.

The protein in cow's milk naturally contains a 4:1 ratio of micellar casein (slow) to whey protein (fast): on paper, it is ideal. Moreover, according to the DIAAS and PDCAAS scales, which both grade proteins based on their bioavailabilities and amino acid profiles, milk protein is indeed superior to whey protein.

Figure 16: Proteins ranked by bioavailability and amino acid profile



Source: Shane Rutherfurd et al. *J Nutr*. 2015 Feb

A good number of studies have compared milk protein with soy protein, or soy-dairy protein blends with whey protein, but not many have compared milk protein with whey protein. In one such study, 20 grams of each led to similar increases in MPS over the 3.5-hour measurement period.[160] In another, 20 grams of each taken twice daily for 12 weeks led to similar increases in muscle mass and strength, when combined with resistance training.[161]

Drinking your own 1:1 blend of whey protein and micellar casein, you can expect an increase in serum EAAs, including leucine, that is as fast as whey's but lasts several hours longer.[162]

Finally, a double-blind randomized controlled trial of resistance-trained men undergoing a supervised 9-week training program found similar increases in strength and lean mass between groups supplementing with either 20 grams of whey protein, 10 grams of whey protein with 10 grams of micellar casein (1:1 ratio), or 4 grams of whey protein with 16 grams of micellar casein (1:4 ratio — the same ratio as in milk).[163]

So it appears that a 4:1 casein:whey blend, a 1:1 casein:whey blend, and whey protein alone exert similar benefits on muscle mass and strength. However, we've seen that, whether in the morning or near bedtime, the fast-digesting whey protein seems a better choice for increasing strength and muscle mass than the slow-digesting micellar casein.

We can tentatively conclude that, though digestion speed does matter, you don't need a lot of fast-digesting protein to maximize MPS. As long as your protein blend contains enough whey, the quality of the rest of the protein still matters, but not its digestion speed. It means that you could take milk protein instead of whey protein — but it also means that taking milk protein, or any complicated "time-release" blend of different proteins, has no benefit over taking just whey protein.

Milk protein has a 4:1 ratio of micellar casein to whey protein. Milk protein, whey protein, and a 1:1 blend of micellar casein and whey protein lead to similar increases in muscle mass and strength. It appears that, given proteins of similar quality, a blend of slow- and fast-digesting proteins won't benefit your muscles more than just a fast-digesting protein.

A look at other ingredients

If you enjoy pure, unflavored whey protein, then by all means, keep doing your thing. However, companies usually add ingredients to give their product a marketing edge (such as a better flavor), so it's worth considering if any of these additives should be sought out — or avoided.

Preservatives

Food preservation covers the use of physical and chemical methods to inhibit microbial growth and retain nutritional quality over time, thereby preventing or slowing decomposition. Traditional methods involved manipulating a food's temperature (boiling, freezing) or physical state (drying, fermentation) or applying natural chemicals (sugar, salt …). Often, these methods were combined into processes, such as curing (drying, smoking, and salting).

Today, these methods are still used, though often with a modern touch. For instance, pasteurization has replaced boiling, but both involve heating; spray-, freeze-, and vacuum-drying are modern methods of dehydration; and artificial preservatives have superseded sugar and salt. Advances in food technology have also led to novel methods of food preservation, such as irradiation.

Protein powders are preserved through drying, as dehydration (removal of the water content) inhibits microbial growth. It is therefore uncommon for protein powders to contain preservatives, be they natural or artificial. Plus, many preservatives cannot legally be used in protein powders (US regulations state not only which preservatives can be used, but in which foods a specific preservative can be used; if a type of food isn't listed, it is excluded by default). The preservatives you may encounter include notably vitamin C (ascorbic acid or ascorbate), vitamin E (tocopherol), and sorbates (calcium, potassium, or sodium sorbate).

Protein powder is preserved through drying, as dehydration prevents microbial growth. The addition of preservatives is therefore uncommon.

Anticaking agents

Anticaking agents are food additives added to powders to prevent clumping (caking). They work either by absorbing moisture or by coating particles to make them water repellent.

Some common anticaking agents include magnesium stearate, silicon dioxide, calcium silicate, tricalcium phosphate, and stearic acid. You may even see powdered rice used.[164]

Most anticaking agents are natural products with well-established metabolic fates (meaning that what happens to them after ingestion is well documented). Magnesium stearate, for example, is simply a combination of magnesium (an essential mineral) and stearic acid (a saturated fatty acid). Calcium silicate is a combination of calcium (an essential mineral) and silica (a trace mineral). At food-additive doses, there is no risk of harm.[165]

A study in some anticaking agents (tricalcium phosphate, calcium silicate, calcium stearate, corn starch, and silicon dioxide) found they hasten the degradation of vitamin C powder in high humidity (>75%),[166] but vitamin C is known to degrade in the presence of water, whereas protein powders are not.

Anticaking agents do not pose a health concern; their addition to protein powders can be ignored.

Soy lecithin

Because no one likes a clumpy protein shake, many whey protein powders contain lecithin, a natural emulsifier that helps the whey protein dissolve in liquids. Lecithin can be found in every cell in your body.[167] The different types of lecithin are composed of various phospholipids, such as *phosphatidylcholine* (PC), *phosphatidylethanolamine* (PE), and *phosphatidylinositol* (PI).

It has been known for decades that dietary lecithin, within the normal diet or as a supplement, gets incorporated in cell membranes and has beneficial health effects on the cardiovascular, nervous, and immune systems.[168] But the amounts in food and supplements are far greater than those found in whey protein powders using lecithin as emulsifier (150–300 milligrams per 30 grams of protein powder, typically: a 0.5–1% concentration).

Lecithin was first identified in egg yolks (and named after them) and has since been found in a variety of foods, with the most common sources today being soybeans and sunflower seeds. Soy lecithin is what you’re most likely to find in whey protein powders, but there is no shortage of articles demonizing it as the worst thing since trans fats simply because it is derived from soybeans.

First, consuming a little soy lecithin as an additive is very different from drinking 3 quarts (2.8 liters) of soy milk per day, as was doing a 60-year-old man when he started suffering from erectile dysfunction, decreased libido, and gynecomastia (an enlargement of breast tissue in men).[169]

Second, most negative perceptions about soy are false, including the idea that regular consumption decreases testosterone and interferes with thyroid function.

Third, soy lecithin oil is nearly 100% fat; it contains very little residual protein and isoflavones (a.k.a. phytoestrogens), the two components that are believed to be implicated in most of soy's purported negative effects on health. You may have heard that a study "found soy lecithin to be strongly estrogenic", but its own data hardly support such a strong conclusion.[170] Having found no trace of genistein (soy's main isoflavone), the authors went to assume that soy lecithin contains "a so-far unidentified estrogen-like compound". Not only that, but they found estrogenic activity in 3 out of 5 infant formulas — one of the formulas with soy lecithin had no estrogenic activity, and one of the formulas with estrogenic activity had no soy lecithin.

There are also many claims about soy lecithin retaining nasty chemicals supposedly used in its production. Such claims don't provide credible sources, when they provide any sources at all. As it stands, soy lecithin production is pretty straightforward: soy oil is degummed, which simply means it is mixed with water in order to partly separate its lecithin component, then this component is dried into a powder.[171]

Finally, some people don't have anything against soy lecithin unless it is sourced from *genetically modified* (GMO) soy. Ignoring the GMO safety debate, soy lecithin is so far removed from soybeans that it contains little to no genetic material and can't be traced back to the soybeans from which it came.[172] So any concerns over GMOs are irrelevant to soy lecithin.

Really, the only *potential* concern with soy lecithin is allergy. Soy protein is a common allergen, but as we said, commercial soy lecithin contains very little residual protein: a mere 100–1,400 parts per million[173] (according to the European Lecithin Manufacturers Association, deoiled soya lecithin is just 0.000065–0.00048% protein[174]). Tested against the immune cells of adults with a soy allergy, soy lecithin caused little to no reaction.[175]

Only two isolated case reports document allergic responses, both in toddlers: one toddler got an allergic response from an allergy test (100 mg of soy lecithin),[176] the other from infant formulas (the three formulas mentioned were 1%, 0.9%, and 0.56% soy lecithin; the Codex Alimentarius limits the lecithin content of formulas to 500 mg per 100 ml of prepared beverage).[177]

Overall, soy lecithin used as a food additive contributes such a minuscule amount of protein that it is generally considered safe for people with soy allergies. That said, everyone is different: if, for any reason, soy lecithin really doesn't agree with you, then avoid it. Just know that most people don't have to.

Some whey protein powders contain lecithin to help the whey protein dissolve in liquids. The lecithin used is usually soy lecithin, but in amounts so small that only people with supersensitive soy allergies might react to it. Most people with soy allergies won't have a reaction.

Thickeners

Some protein powders include thickening agents to create a creamier shake. Approved thickeners include starches (corn, potato, tapioca …), gums (xanthan, guar, locust bean …), and sugar polymers (pectin, agar, carrageenan …).

There isn't much to say about the starches because they are likely used in quantities too low to have a notable nutritional impact. At most, they might add a gram or two of carbs per serving.

Gums are also not much of a health concern. In large amounts, they act as soluble fibers: they bind to water, increasing viscosity and slowing digestion, and can thus lower post-meal blood-sugar response if the meal contained carbs.[178] But the doses used in protein powders are way too small to have any noticeable effect.

The one thickener that should give you pause is carrageenan. Although it has been granted *generally recognized as safe* (GRAS) status by the FDA, there are still gaps in our understanding of this sugar polymer.[179] Toxicological reviews deem it safe at incredibly high doses,[180] around 18–40 milligrams per kilogram of body weight (mg/kg),[181] but concerns remain about how it interacts with the digestive system when consumed in a solution, such as a whey protein shake, rather than in solid food.[182]

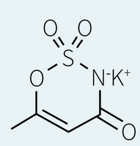
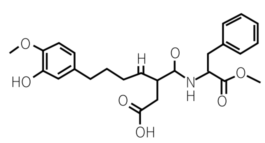
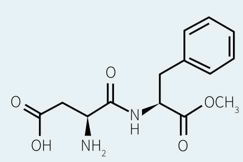
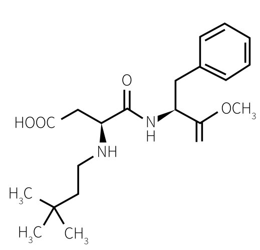
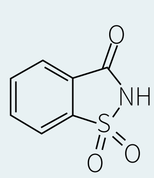
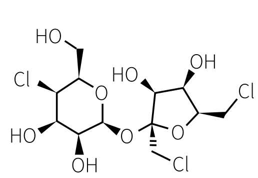
There are reasons to believe that carrageenan may worsen gut problems (e.g., inflammatory bowel disease or irritable bowel syndrome), might harm the gut microbiome, and might promote inflammation in the colon.[179] The amounts of carrageenan used in whey protein powders would likely be low and probably of little concern, but we don't really know. Minimizing exposure may be prudent.

Thickeners are safe, except maybe for carrageenan, which might have negative effects on gut health and warrants caution.

Artificial sweeteners

Artificial sweeteners are synthetic sugar substitutes that are many times sweeter than sugar but have little to no caloric value and generally do not affect blood sugar. There are currently six FDA-approved artificial sweeteners:[183] acesulfame potassium (Ace-K), advantame, aspartame, neotame, saccharin, and sucralose.

Table 6: FDA-approved artificial sweeteners

Sweetener	Brand-name examples	Sweetness relative to table sugar (sucrose)	Acceptable Daily Intake (ADI) in mg/kg/day*	Tabletop sweetener packets to reach the ADI**
Acesulfame potassium (Ace-K) 	Sweet One® Sunett®	200x	15	23
Advantame 	None yet	20,000x	32.8	4,920
Aspartame 	Equal®	200x	50	75
Neotame 	Newtame®	7,000–13,000x	0.3	16–30
Saccharin 	Sugar Twin® Sweet'N Low® Necta Sweet®	200–700x	15	22–79
Sucralose 	Splenda®	600x	5	23

*milligrams per kilogram of body weight per day | **based on a 60 kg (132 lb) person

An in-depth discussion on artificial sweeteners is beyond the scope of this guide, but we do want to address some of the common areas of controversy, starting with general safety.

The FDA has set an Acceptable Daily Intake (ADI) for each artificial sweetener after evaluation of the chemical's toxic and cancer-causing effects.[183] Depending on how much protein powder you consume and how much sweetener it contains, the ADI may or may not be something to worry about. Of course, you also need to factor in other foods if they contain the same sweetener.

Unfortunately, manufacturers seldom list the amount of sweetener in a food or supplement. In theory, since ingredients must be listed by weight (from heaviest to lightest), you can get a general idea of how much sweetener a product contains, but only if you can guess the amounts of the surrounding ingredients.

Another issue worth mentioning is the link observational data found between consumption of artificial sweeteners and obesity.[184] Fortunately, it can certainly be explained by reverse causality: it isn't that people who use artificial sweeteners are more likely to become overweight, but that overweight people are more likely to use artificial sweeteners (in an attempt to lose weight).[185] As it stands, intervention studies have consistently shown that artificial sweeteners do not cause weight gain;[186] on the contrary, they commonly reduce energy intake and promote weight loss.[187]

The one exception appears to be saccharin, which was recently shown to promote weight gain to the same extent as table sugar over 12 weeks, while sucralose, aspartame, and stevia did not.[188] All five of the sweeteners were consumed in a beverage in amounts within the acceptable daily intake limits. However, diet wasn't controlled, so it is possible that food intake was higher in the saccharin group than in the sucralose, aspartame, and stevia groups, especially considering that, over the course of the study, hunger was greater in the saccharin group than in the four other groups. In other words, it is possible that saccharin promotes weight gain indirectly by increasing hunger, but this hypothesis would need to be verified in specially designed studies.

Lastly, there are concerns over artificial sweeteners interfering with glycemic control and reducing insulin sensitivity. These concerns seem to stem mainly from studies on sucralose showing that a realistic daily intake of 150–200 mg reduces insulin sensitivity in healthy adults.[189,190] A previous study, however, had found no such effect from a much higher daily intake (1,000 mg),[191] so the data are conflicting and the question remains unresolved.

Frankly, this entire discussion is somewhat moot since finding protein powders void of artificial sweeteners isn't difficult, if that's what you want.

Artificial sweeteners are calorie-free synthetic sugar substitutes. There is no reason to believe they are harmful, in reasonable doses, and they are unlikely to promote weight gain or glucose intolerance.

Natural nonnutritive sweeteners

Natural nonnutritive sweeteners are naturally occurring sugar substitutes that are many times sweeter than sugar but have little to no caloric value and generally do not affect blood sugar. The FDA has granted *generally recognized as safe* (GRAS) status to two such sweeteners: steviol glycosides, from the leaves of the *Stevia rebaudiana* plant, and mogrosides, from *Siraitia grosvenorii* (*luo han guo*, or monk fruit).[183]

Table 7: FDA-approved natural nonnutritive sweeteners

Sweetener	Brand-name examples	Sweetness relative to table sugar (sucrose)	Acceptable Daily Intake (ADI) in mg/kg/day*	Tabletop sweetener packets to reach the ADI**
Monk fruit extract	Monk Fruit in the Raw® PureLo®	100–250x	Not determined	Not determined
Stevia extract	Truvia® Pure Via™ Enliten®	200–400x	10–12	20

*milligrams per kilogram of body weight per day (d) | **based on a 60 kg (132 lb) person

Importantly, the FDA approved only stevia extracts that are more than 95% steviol glycosides. Stevia leaves and crude stevia extracts are not GRAS; they cannot be sold as sweeteners in the US. This is important because stevia's adverse effects, such as a decreased testosterone, are linked to the stevia leaf, not to steviol glycosides.[192]

Steviol glycosides include notably Rebaudioside A (also known as Reb A) and stevioside. They are resistant to your digestive enzymes;[193] they pass intact through your gastrointestinal tract, breaking down only after coming into contact with your colon's microbiome.[194] The microbes (the bacteria) remove and metabolize the sugar molecules from the steviol backbone, which is then absorbed into your blood, metabolized within your liver, and excreted in your urine.[195]

Far less research has focused on the monk fruit and its sweet-tasting mogrosides.[196] Although monk fruit extracts have been granted GRAS status by the FDA, and have long been used in traditional Chinese medicine, further research is necessary to determine their potential health effects and safe upper intake levels.

Natural nonnutritive sweeteners are calorie-free natural sugar substitutes. Stevia extracts that are more than 95% steviol glycosides are safe, whereas stevia leaves and crude stevia extracts are not. Monk fruit extracts appear to be safe, but research on their effects is still scarce.

Polyols

Polyols (sugar alcohols) are another class of sweeteners (sugar substitutes).[197] The six polyols most used as sweeteners are erythritol, lactitol, maltitol, mannitol, sorbitol, and xylitol; compared to sugar, they are 30–100% as sweet, lower in kilocalories (0.2–2.7 per gram, instead of 4), and lower on the Glycemic Index (meaning they have a lesser effect on blood sugar).

Table 8: Most common polyols

Sweetener	Glycemic index*	Calories (kcal/g)	Sweetness**
Erythritol	0	0.2	0.6–0.8
Lactitol	6	1.9	0.3–0.4
Maltitol	35	2.1	0.9
Mannitol	0	1.6	0.5–0.7
Sorbitol	9	2.7	0.5–0.7
Xylitol	13	2.4	1.0

*glucose = 100; sucrose = 65 | **relative to sucrose; sucrose = 1.0

Except for erythritol,[198] polyols may cause bloating and diarrhea when consumed in excess, since they are only partially absorbed in the gastrointestinal tract and are rapidly metabolized by the microbiome in the colon.[199]

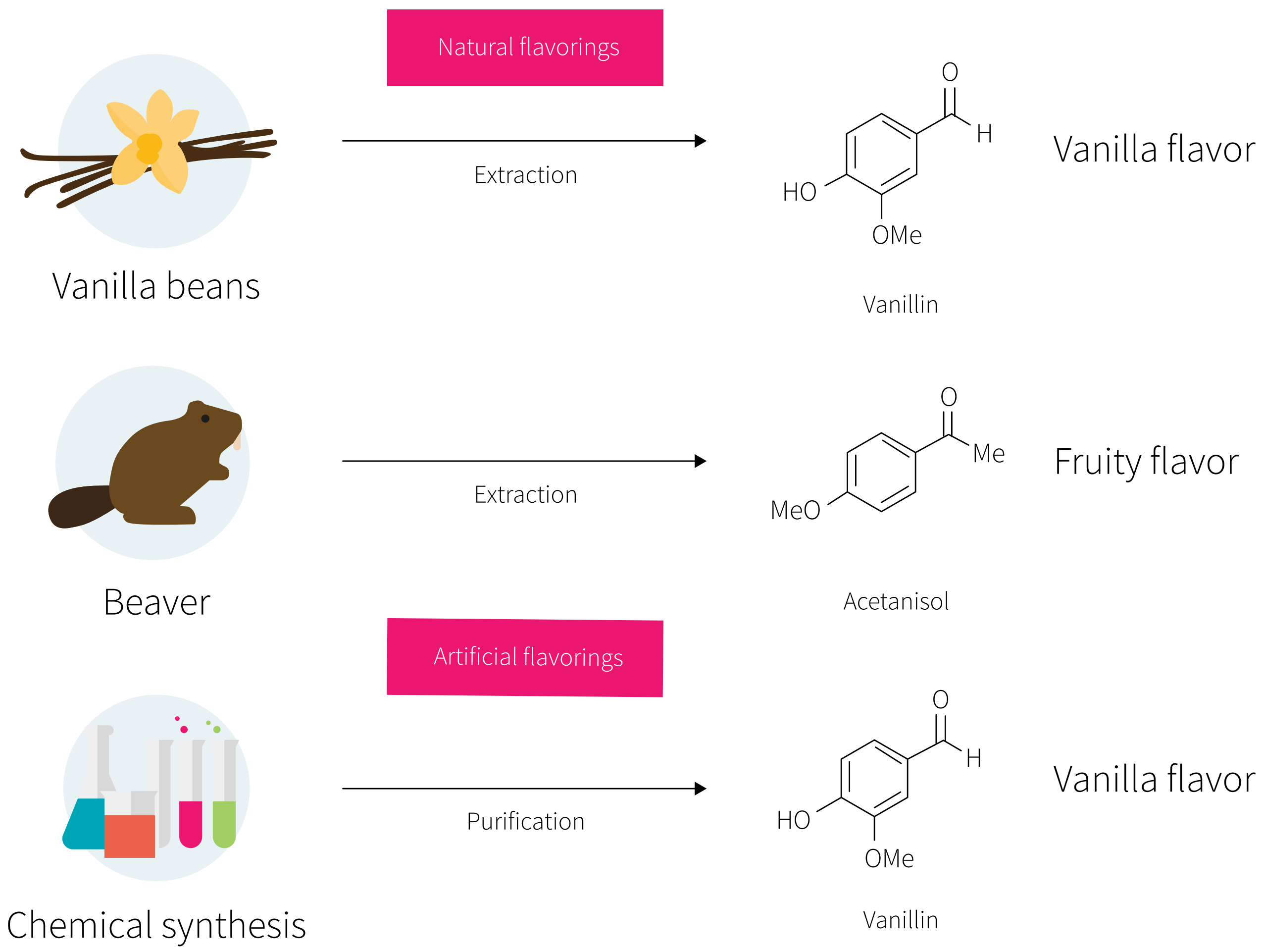
Preliminary evidence suggests that erythritol and, to a lesser extent, xylitol may help prevent dental plaque and cavities,[200,201] but more studies are needed for confirmation and to determine an optimal protocol — amount, frequency, and exposure time (relevant studies use polyols mostly in chewing gums or hard candies, to ensure prolonged dental exposure).

Compared to sugar, the six polyols most commonly used as sweeteners are 30–100% as sweet, are lower in calories, and have a lesser effect on blood sugar. Except for erythritol, they may cause bloating and diarrhea when consumed in excess.

Natural and artificial flavorings

Protein powders come in all kinds of flavors. Historically, **natural flavorings** were called extracts, tinctures, or essential oils; most are isolated from plants. **Artificial flavorings** are synthesized in a lab; most contain the exact same molecules that exist naturally in foods or are formed during food preparation, but some molecules are only structurally similar.

Figure 17: How natural and artificial flavorings are obtained



A flavoring is usually a combination of more than 50 molecules. Rarely does a flavor originate in a single molecule, as in the case of vanilla (vanillin), strawberry (ethyl methylphenylglycidate), or green apple (hexyl acetate).

Contrary to popular belief, artificial flavorings are probably safer than natural ones, which are more likely to vary in their composition and to contain impurities.[202] A natural vanilla extract, for instance, is a mixture of several hundred different molecules in addition to vanillin.

Still, whether natural or artificial, the thousands of flavorings used by the food industry are *generally recognized as safe* (GRAS). To be granted GRAS status, a flavoring molecule must undergo evaluation of its (1) chemical structure and physicochemical properties, (2) purity and manufacturing process, (3) natural occurrence in foods, (4) potential exposure level, (5) metabolism, (6) toxicology, and (7) gene-damaging potential.[203]

If information for criteria 5–7 is not available, however, a flavoring, whether natural or artificial, may still be granted GRAS status based on the structural similarity of its molecules to other, tested molecules. It means that GRAS status can be granted to (1) untested molecules and (2) untested combinations of molecules (the actual flavorings).

Furthermore, flavorings are granted GRAS status by the FDA based on assessments by a scientifically independent Expert Panel funded by the Flavor and Extract Manufacturers Association.[204] In other words, while the FDA has the last say, it does not assess the flavorings itself. In the past, it has banned several artificial flavorings only after being petitioned for their removal from the food supply due to some animal studies suggesting their being carcinogenic.

This serves to show that flavorings can pose health issues and that their safety evaluation isn't always thorough. The sheer number of chemicals used as flavorings makes testing each and every one a daunting task, and since most are used only rarely, there is little economic incentive to support a traditional toxicology battery (meaning that too many flavorings are granted GRAS status based on structural similarities, as explained above).

Thousands of natural and artificial flavorings have been deemed safe by the FDA based on external assessments by an industry-funded committee. Alas, the evaluation process isn't always thorough.

Natural and artificial colorants

Without food colorants, all whey protein powders would share a similar off-white color, regardless of the included flavorings.

Seemingly innocuous, food colorants are probably the most rigorously regulated food additives in the world. Unlike flavorings, they can't simply be granted GRAS status; they are assessed by the FDA directly, so the safety evaluation they undergo is much stricter.

As with flavorings, there are two general categories of food colorants: natural and artificial.

Natural colorants are derived from natural sources, mostly plants; they include β-carotene, annatto, paprika, turmeric, and beet powder, among many others. They're considered safe and are seldom controversial (one exception being E120, a red colorant derived from an insect, the cochineal).

Conversely, **artificial colorants**, or food dyes, are often controversial, notably because different countries using different approval methods.[205] For example, of 18 food dyes approved by either the FDA (in the US) or the EFSA (in the EU), only 6 are approved by both agencies.

Table 9: Food dyes approved in the US and EU

Food dye	US	EU
FD&C Yellow No. 5 (Tartazine; E102)	YES	YES
FD&C Yellow No. 6 (Sunset Yellow; E110)	YES	YES
FD&C Red No. 3 (Erythrosine; E127)	YES	YES
FD&C Red No. 40 (Allura Red AC; E129)	YES	YES
FD&C Blue No. 1 (Brilliant Blue FCF; E133)	YES	YES
FD&C Blue No. 2 (Indigotine; E132)	YES	YES
FD&C Green No. 3 (Fast Green FCF)	YES	NO
Orange B	YES	NO
Citrus Red No. 2	YES	NO
FD&C Red no. 2 (Amaranth; E123)	NO	YES
Quinoline Yellow (E104)	NO	YES
Carmoisine (E122)	NO	YES
Cochineal Red A (E124)	NO	YES
Patent Blue V (E131)	NO	YES
Green S (E142)	NO	YES
Brilliant Black PN (E151)	NO	YES
Brown HT (E155)	NO	YES
Lithol Rubine BK (E180)	NO	YES

Reference: Lehto et al. Food Addit Contam Part A Chem Anal Control Expo Risk Assess. 2017 Mar.

A notable topic of controversy, the effect of food dyes on *attention-deficit hyperactivity disorder* (ADHD) in children was heavily investigated in the ’70s and ’80s. The most recent meta-analysis dates from 2012: it includes 20 double-blind randomized controlled trials and concludes that food dyes slightly promote hyperactivity in children.[206] The authors speculate that 8% of children with ADHD may benefit from eliminating dyes from their diets.

Still, the association between food dyes and ADHD isn’t entirely clear, for at least three reasons.[207] **First,** it hasn’t been investigated in adults. **Second,** food dyes appear to worsen

ADHD only in children with certain genes.[208] **Third,** most studies have used a dye mixture, leaving open the possibility that only some dyes worsen ADHD.

Aside from potentially worsening ADHD in children, food dyes approved in the US have been shown to be carcinogenic or genotoxic (damaging to genetic material) to varying extents, and they can also cause hypersensitivities and allergic reactions in susceptible individuals.[209] For example, Blue #1, Red #40, Yellow #5, and Yellow #6 can cause allergic reactions, and both yellow dyes also contain benzidine, a carcinogen.[210,211,212]

Table 10: Genotoxic potential of seven food dies

	Studies positive for genotoxicity	Studies negative for genotoxicity
FD&C Yellow No. 5 (Tartazine; E102)	6	5
FD&C Yellow No. 6 (Sunset Yellow; E110)	2	8
FD&C Red No. 3 (Erythrosine; E127)	4	8
FD&C Red No. 40 (Allura Red AC; E129)	3	7
FD&C Blue No. 1 (Brilliant Blue FCF; E133)	2	7
FD&C Blue No. 2 (Indigotine; E132)	1	10
FD&C Green No. 3 (Fast Green FCF)	3	6

Reference: Kobylewski & Jacobson. Int J Occup Environ Health. 2012 Jul–Sep.

These findings suggest that the safety evaluation of food dyes, though stricter than the safety evaluation of flavorings, is still inadequate. Many of the studies the FDA used to declare food dyes safe were conducted by dye manufacturers, were too short to adequately assess carcinogenicity, and tested each dye only in isolation (whereas dyes are often combined).

Natural colorants are not a concern, but there are data linking some artificial colorants (food dyes) to cancer, genotoxicity, allergies, and behavioral alterations. While nothing conclusive can be drawn from the available research, it seems prudent to avoid food dyes whenever possible.

Digestive enzymes

Proteases are digestive enzymes that break down proteins. They are used in some protein powders to increase the bioavailability of the protein. Two studies have investigated this topic, both completed by the same research lab and funded by Triarco, the company that patented the tested product (Aminogen®, a blend of proteases isolated from the fungi *Aspergillus niger* and *Aspergillus oryzae*).

The first study assessed how adding Aminogen® to whey protein concentrate affected serum levels of amino acids during the 4 hours following ingestion.[213] Two groups of 21 healthy young men took 50 grams of whey protein twice: first on its own, then, nine days later, with Aminogen® — 2.5 grams for one group, 5 grams for the other. Both doses resulted in similarly higher serum levels of amino acids, suggesting that the benefit tops out at 2.5 grams or less.

The second study was a randomized controlled trial involving 36 resistance-trained men who took 40 grams of whey protein concentrate twice daily (80 g/day) for the 4 weeks of a resistance-training program.[214] One group of 18 men took only whey protein; the other took whey protein blended with 1.5 grams of Aminogen® (3 g/day). Minimal differences in clinical markers of cardiometabolic health and liver and kidney functions were seen between the groups.

Neither study tells us much. The effects are small to nonexistent, and useful parameters such as MPS and lean body mass were not assessed. Plus, while there are many types of digestive enzymes, only a single, specific blend of two was tested. At this time, there isn't strong evidence supporting the use of digestive enzymes in whey protein powders, although this area of investigation remains largely unexplored.

On a final note, even if Aminogen® had been proven to make a practical difference, dosage would still be an issue. At least one lawsuit was filed against a company for using less than 0.1% of Aminogen® in their whey protein product; the company lost the case because the above studies used concentrations 36–91 times greater (3.6–9.1%).

There is currently no evidence that adding digestive enzymes to whey protein has any practical benefit. There are only two studies on this topic; both used the same patented enzymatic blend, and neither provided data on body composition or performance outcomes.

BCAAs

Whey protein naturally contains *branched-chain amino acids* (BCAAs), but people also take BCAAs separately to build muscle or, if they eat below maintenance or train fasted, to prevent muscle loss. Since whey protein can serve both purposes, two questions spring to mind:

1. Is one supplement better than the other for either purpose?

2. Are the two supplements redundant or complementary?

We'll answer both questions in detail, but first, let's review some background information.

All protein, including the protein you eat and the protein in your body, is made from some combination of 20 amino acids. The 11 your body can synthesize are called *nonessential amino acids* (NEAAs). The 9 your body cannot synthesize — and thus needs to get from food — are called *essential amino acids* (EAAs).

There are times, however, when specific NEAAs may become essential due to a disease, temporary illness, or some other condition. For example, burn-victim bodies use glutamine faster than they can make it.[215] Such amino acids are called *conditionally essential*.

Table 11: Classification of amino acids by essentiality

Essential (EAA)	Nonessential (NEAA)	Conditionally essential
|| **Leucine** || || **Isoleucine** || || **Valine** || Histidine Lysine Methionine Phenylalanine Threonine Trytophan	Alanine Asparagine Aspartic acid Glutamic acid Serine	Arginine Cysteine Glutamine Glycine Proline Tyrosine

|| **BCAA** ||

There are also amino acids, such as β-alanine and taurine, that aren't used in protein synthesis but still play a role in metabolism. These *nonprotein amino acids* are covered in the next section.

Muscle building requires that, on average, *muscle protein synthesis* (MPS) exceed *muscle protein breakdown* (MPB), resulting in a net accumulation of muscle protein. All 20 amino acids are needed to build muscle tissue,[216] but MPS is stimulated primarily by the EAAs,[217] and among the EAAs by the BCAAs,[218,219] and among the BCAAs by leucine.[220] (As we saw, whey is high in EAAs, notably leucine.)

Digging Deeper: Why MPS matters

Muscle protein synthesis (MPS) is the process of building skeletal muscle tissue, whereas *muscle protein breakdown* (MPB) is the process of breaking down skeletal muscle tissue. MPB is necessary for muscle growth and adaptation,[221] but for your muscle mass to increase, you need your MPS to exceed your MPB (overall, in the long term).

Note that "protein synthesis" refers to the creation of any protein in your body. If your interest lies in muscle growth, you need to focus on MPS and MPB specifically, lest you be misled by changes in whole-body protein synthesis of the liver, kidneys, and gut.

In fact, you should focus on MPS.

When you eat, you produce insulin. Insulin plays only a minor role in stimulating MPS, and only a little insulin (≈15 IU/mL) is necessary to suppress MPB.[222] Consequently, MPS and MPB are affected similarly by 25 grams of protein and by a combination of 25 grams of protein and 50 grams of carbs, even though the latter causes a fivefold greater insulin response.[223]

Since you need very little protein to reduce MPB, adding more protein only serves to increase MPS.[224] But then, does MPS translate into actual muscle growth?

The first study to investigate this question answered in the negative.[225] The major problem with this study is that it measured MPS only once, for 6 hours, even though bouts of resistance training affect MPS for 1–3 days.[226] More precisely, MPS was measured first at rest then after exercise, but only at the onset of 16 weeks of resistance training. Those MPS measurements (taken on the first day) were later compared to the muscle-gain measurements (taken on the last day).

Two subsequent studies overcame this limitation by analyzing MPS around the clock.[227,228] They did find a correlation between muscle growth and increases in MPS, but only after 3 weeks, probably because excessive muscle damage in the early stages of lifting overshadowed the correlation.

So, yes, long-term increases in MPS lead to muscle growth. Alas, long-term increases in MPS (days to weeks) are difficult to estimate from short-term increases in MPS (minutes to hours).

There are 20 amino acids involved in protein synthesis. The 9 your body cannot synthesize and thus must get through food are called *essential amino acids* (EAAs). *Muscle protein synthesis* (MPS) is stimulated primarily by the EAAs, and among the EAAs by the *branched-chain amino acids* (BCAAs), and among the BCAAs by leucine.

Can BCAAs alone build muscle?

Technically, no, since your muscles are composed of all 20 amino acids. In practice, BCAAs taken alone can promote muscle growth — if your body can get the 17 other amino acids in some other way (it can synthesize 11 of them; the other 6 it may find in some food you're still digesting, for instance). Still, BCAAs or even EAAs taken alone stimulate MPS less than the same amount of BCAAs or EAAs from whey protein.[219,229]

But isn't leucine the most anabolic of the amino acids? It is, and yet, taken alone in a fasted state, it increases MPS and anabolic signaling (notably through the mTOR/p70S6K pathway) for about 1.5 hours only.[220,230,231] MPS stops as soon as another of the EAAs gets depleted. BCAAs contain only 3 of the 9 EAAs: leucine, isoleucine, and valine.

One study compared 25 grams of whey protein (providing 3 grams of leucine), 6.25 grams of whey protein mixed with leucine (3 grams of leucine in total), and 6.25 grams of whey protein mixed with EAAs (0.75 grams of leucine in total, and as much of the other EAAs as in 25 grams of whey protein).[232] At the 3-hour mark post-fasted-exercise, all three supplements stimulated MPS similarly, but at the 5-hour mark, only the pure whey protein still stimulated MPS. Without the exercise stimulus, however, even the pure whey protein could not increase MPS past the 3-hour mark.

All right. But it isn't too surprising that, for a same amount of leucine, the pure whey protein (which contains more amino acids total) should win out. What would happen, though, if you increased the dose of added leucine? A follow-up study set to answer that question.[233] It found that, at the 4.5-hour mark post-fasted-exercise, 25 grams of whey protein (providing 3 grams of leucine) and 6.25 grams of whey protein plus 4.25 grams of leucine (5 grams of leucine in total) stimulated MPS similarly, whereas 6.25 grams of whey protein plus 2.25 grams of leucine (3 grams of leucine in total) no longer stimulated MPS.

Interestingly, this same study also found that 6.25 grams of whey protein mixed with BCAAs (5 grams of leucine in total) stimulated MPS less than 6.25 grams of whey protein plus 4.25 grams of leucine (also 5 grams of leucine in total). In other words, leucine stimulated MPS more when *not* taken alongside the two other BCAAs, possibly because all three BCAAs share intestinal and muscular transporters, so that isoleucine and valine compete with leucine for both absorption in the gut and entry into muscle tissue.[234,235]

If you don't get enough protein, you can take small doses of leucine to compensate — to some extent, and only with regard to muscle building. Importantly, BCAAs or EAAs taken alone stimulate MPS less than the same amount of BCAAs or EAAs from whey protein, which contains all 20 amino acids.

Can BCAAs alone stop muscle loss?

BCAAs can help *slow* muscle loss, but they cannot stop or prevent it.

The *amino acid pool* represents all the amino acids available to your body for protein synthesis and other functions. Whenever your body uses amino acids, it gets them from this pool, which gets replenished by the protein you eat and through breakdown of your body's own protein.

Your body is constantly breaking down its old and damaged proteins, recycling any amino acids it can, and rebuilding the broken-down proteins if appropriate, in a process called *protein turnover*.

There are 20 amino acids. Your body can synthesize the 11 NEAAs, but, when you're in a fasted state, it can only get the EAAs it needs through the breakdown of its own protein. Following an overnight fast, about 85% of the protein your body is ready to sacrifice comes from its skeletal muscle,[236] and MPB is about 30% greater than MPS — your body is releasing EAAs from your skeletal muscle to synthesize the proteins your organs need to keep you alive.[237]

Some MPB is necessary for muscle growth and adaptation.[221] Exercise increases MPS, but also MPB, leaving your body in a catabolic state.[238] After exercise, consuming of all 9 EAAs can ensure that your body won't need to keep breaking down its own proteins; it will shift into an anabolic state by suppressing MPB and increasing MPS.[239]

However, consuming only some of the EAAs — only the BCAAs, for instance, or only leucine — won't have the same effect. Following an overnight fast, BCAA infusion suppressed both MPB *and MPS* in two studies, one lasting 3 hours and the other 16 hours.[240,241] Because MPB suppression is stronger, overall net protein loss is slightly reduced, but the body remains in a catabolic state throughout.

This makes sense, considering that leucine provides only a temporary increase in MPS by using up any available amino acids in the amino acid pool. When the available EAAs are used up, MPS returns to fasted baseline, all the faster since the simultaneous reduction in MPB further reduces the amount of EAAs available.

When you're in a fasted state, taking BCAAs can only slow muscle loss, whereas taking a protein rich in all the EAAs can stop it.

Do BCAAs benefit body composition?

As we've just seen, BCAAs alone cannot promote muscle growth; but what if you add them to your daily food? Leucine especially: since it increases anabolic signaling and MPS, cannot it promote muscle growth if your diet contains the required building blocks?

In healthy, untrained women performing regular resistance-training workouts, taking 10 grams of EAAs (providing 5 grams of BCAAs, including 2 grams of leucine) both before and after a training session, as well as in the morning on non-training days, didn't alter body composition.[242]

Likewise, in healthy, untrained men performing regular resistance-training workouts, taking 4.5 grams of BCAAs (providing 2.25 grams of leucine) both before and after a training session (four sessions per week) didn't alter body composition.[243]

On the other hand, during a 21-day trek through the Andes, 5 grams of BCAAs (providing 2.5 grams of leucine) taken thrice daily appeared to preserve muscle mass; but we don't know how much protein the participants were eating.[244]

Likewise, in resistance-trained men on a hypocaloric high-protein diet (2.7 g/kg/day), taking 7 grams of BCAAs (providing 3.5 grams of leucine) before/during and after a training session was reported to preserve muscle mass.[245] However, other researchers have criticized the study for shady data reporting leading to a flawed conclusion.[246]

Another study in extremely lean (6–8% body fat) elite wrestlers found that BCAAs didn't help preserve muscle mass but did increase fat loss during a 19-day aggressively hypocaloric diet.[247] However, protein intake was low (1.1 g/kg) and the BCAA dose unrealistic: 0.9 g/kg/day (roughly 67 grams per day, on average, providing some 50 grams of leucine).

Table 12: Effect of BCAA supplementation on the body composition of healthy young adults

Year of publication	Duration	N	Training status	Daily dose	Fat-free mass	Fat mass
1992	21 days	13	Recreationally active	15 g BCAAs (7.5 g leucine)	↑	↔
1997	19 days	31	Elite wrestlers	67 g BCAAs (50 g leucine)	↔	↓
2011	8 weeks	19	Untrained	9 g BCAAs (4.5 g leucine)	↔	↔
2012	8 weeks	17	Resistance trained	28 g BCAAs (14 g leucine)	↑	↓

In healthy young adults, BCAAs don't seem to promote muscle growth, but when combined with exercise and a hypocaloric diet, they may help reduce muscle loss or increase fat loss.

A meta-analysis of studies in older adults showed that leucine supplementation increased MPS but not lean body mass.[248]

Digging Deeper: Why doesn't more MPS always translate into more muscle?

As we saw, *muscle protein synthesis* (MPS) appears to translate into muscle growth in the long term. Yet studies don't always report such a result. Why?

Mostly, because muscle growth is slow. Its detection requires following large groups of people for at least several months, to make up for the **imperfect accuracy** of most body-composition measurements and **low sensitivity** of most body-composition analyses.

The **imperfect accuracy** of measurements is mostly caused by confounding factors. For instance, a small muscle gain could be hidden by a small water loss, because water is part of your lean mass.

For simplicity's sake, we often conflate increases in lean mass with increases in muscle mass, but as a rule, studies don't really measure muscle mass. Instead, they measure two things: fat mass and, by elimination, lean mass. Lean mass is everything that isn't fat mass — including your skeletal muscle, yes, but also your bones, your organs minus their fat, and the water in your blood and cells.

The **low sensitivity** of analyses is why the largest meta-analysis of protein-supplementation studies to date reported a benefit on lean mass and muscle-fiber size — even though 23 of the 26 lean-mass studies and 6 of the 11 muscle-fiber studies reported no statistically significant benefits.[13] It's not that protein supplementation didn't have an effect; each individual study was simply too small and underpowered to detect it. It took a meta-analysis of these studies to obtain enough statistical power to observe the benefit.

MPS studies and body-composition studies can be seen as complementary. The MPS studies are specific to muscle tissue, are better controlled, and give deeper mechanistic insights. The body-composition studies give us an idea of the overall magnitude of effect and of influences outside MPS. When, in the same study, both the MPS measurements and body-composition measurements line up, you can be relatively sure the intervention is working as intended.

Adding 2.5 grams of leucine to three meals (7.5 g/day) did not benefit the muscle mass of healthy elderly men over the course of 3 months,[249] nor that of elderly men with type 2 diabetes over the course of 6 months.[250] Likewise, adding 5 grams of leucine to two meals (10 g/day) did not benefit benefit the body composition of older adults undergoing resistance training for 3 months.[251]

Two studies in the elderly used EAA supplements. Sedentary elderly women saw their muscle mass increase from 15 grams of daily EAAs (providing 2.8 grams of leucine) for 3 months,[252] whereas resistance training increased muscle mass regardless of whether elderly men took a placebo or 12 grams of EAAs (providing 2.2 grams of leucine) postexercise.[253]

Spiking meals with leucine doesn't appear to benefit muscle mass or body composition in the elderly, whether they exercise or not.

Do BCAAs benefit muscular strength, function, or recovery?

Taking 5 grams of BCAAs (providing 2.5 grams of leucine) thrice daily during a 21-day trek through the Andes appeared to preserve both muscle mass and leg strength.[244] Also, although taking 5 grams of leucine twice daily with a meal did not benefit the body composition of older adults undergoing resistance training for 3 months, it did benefit their muscle strength and function (the latter being measured through a standing-balance test, a walking-speed test, etc.).[251]

However, BCAAs didn't affect muscle strength or function in elite wrestlers or untrained adults.[242,243,247] Likewise, adding to meals some 2.5 grams of leucine (alone or with other EAAs) didn't seem to benefit muscle strength or function in sedentary elderly adults,[249,250,252] even when the study included a resistance training program (both the supplement group and the placebo group saw similar improvements from their training).[253]

According to a meta-analysis of eight randomized controlled trials, BCAAs (3–32 g) taken around a training session can reduce perceived muscle soreness the next day.[254] But a different analysis remarked that only low-quality studies found benefits; high-quality studies found none.[255]

BCAAs might reduce muscle soreness when taken around resistance-training sessions, but they don't appear to benefit muscular strength or function.

Do BCAAs help during muscle unloading?

Up to now, we've looked at studies in people free to move around and live their lives. But what if you *can't* move around?

The muscles of a limb in a cast aren't being used, and neither are those of a bedridden body. This total inactivity, called muscle unloading, causes anabolic resistance and reductions in MPS.[256] A mere 7 days of bed rest significantly reduces MPS and muscle mass in young men and elderly adults alike.[257,258] So, during prolonged bed rest, can taking BCAAs help?

Apparently yes, but only when protein intake is rather low (around the RDA of 0.8 g/kg).[259] Studies on higher intake levels (1.2 g/kg and higher) showed no benefit.

For example, in bedridden middle-aged adults with a protein intake of about 0.9 g/kg, adding 4.4 grams of leucine to three meals (13.2 g/day) appears to reduce fat gain and muscle loss, as well as preserve muscle strength and function.[260]

Likewise, in elderly adults with a protein intake of about 0.9 g/kg, 20 grams of EAAs (providing 3.6 grams of leucine) taken twice per day reduced muscle loss during the 6 weeks following surgery, although muscle function was unaffected.[261]

However, in young adults with a protein intake of about 1.3 g/kg, adding 2.5 grams of leucine to three meals (7.5 g/day) did not affect muscle mass or strength during limb immobilization.[262]

Supplementing with EAAs, BCAAs, or leucine during bed rest or limb immobilization may help preserve muscle mass and strength if protein intake is low, but likely not if protein intake is adequate.

BCAAs vs. whey

All of the studies discussed compared BCAAs with nothing or some carbohydrate, rather than with a complete protein. If we are looking for something to add to meals or take after training, we also need to consider protein powders such as whey.

The largest meta-analysis to date has shown that, when combined with resistance training, protein supplementation benefits muscle mass and strength,[13] even in elderly adults,[263] and when only looking at whey protein supplementation.[264] Additionally, a meta-analysis of whey protein studies has shown that taking 25 grams of whey protein before or after exercise reduces muscle soreness and improves recovery for up to 3 days.[265]

The benefits seen from whey protein are both larger and more consistent than the benefits seen from leucine, BCAA, or EAA supplementation. A complete, fast-digesting protein, such as whey, should be your first choice, but if for whatever reason a protein powder is not an option for you, then some isolated leucine, BCAAs, or EAAs may be useful.

Whey protein provides all the benefits of BCAAs and then some.

Nonessential & nonprotein amino acids

In addition to BCAAs and leucine, several *nonessential amino acids* (NEAAs) and *nonprotein amino acids* (NPAAs) are marketed as performance-enhancing or health-promoting supplements. NPAAs are amino acids that play a role in metabolism but are not involved in protein synthesis. There are at least 140 known NPAAs in existence, a few of which will be covered here.[266]

Glutamine

Glutamine (L-glutamine) is the most abundant amino acid in your body. It is an NEAA because your body can usually make some as needed, but it can become essential in some circumstances, such as when physical trauma is exceptionally high. It is *conditionally essential*.[215]

One of glutamine's roles in your body is to help get leucine inside your cells. It does so by entering a cell on its own then leaving it using a transporter that simultaneously pulls in leucine. Basically, when the cell kicks out glutamine, it brings in leucine. This process is necessary for the stimulation of *mammalian target of rapamycin* (mTOR, one of the main anabolic pathways) and protein synthesis.[267]

The prominent role played by glutamine in amino acid transport and protein synthesis brings up the question of whether glutamine supplementation can enhance muscle growth or exercise performance.

A handful of studies have investigated the effects of glutamine supplementation on body composition, and a meta-analysis of these studies found no benefit.[268] Even the study using the highest dosage of 0.9 g/kg/day in resistance-trained adults found no effect.[269] There may be a benefit to exercise recovery,[270] especially when glutamine is combined with leucine,[271] but more research is needed for confirmation.

Endurance athletes who train a lot may benefit in another way, though. Glutamine plays an important role in immune function (it is notably the primary fuel source of white blood cells).[272] After prolonged endurance exercise, plasma glutamine levels are reduced, which correlates with an increased risk of infection.[273] Glutamine supplementation should help prevent or lessen this increase.

Relatedly, prolonged endurance exercise is known to cause leaky gut, a condition in which heat stress and reduced blood flow to the gastrointestinal tract cause intestinal cell damage.[274] This damage loosens tight junctions between cells, allowing for the absorption of things that are not supposed to pass through the intestinal barrier (e.g., proinflammatory endotoxins). Glutamine supplementation reduces exercise-induced intestinal permeability and the resulting increase in serum endotoxin and inflammatory markers.[275,276]

Importantly, at least one study in patients with Crohn's disease (a type of inflammatory bowel disease) has reported that glutamine and whey protein similarly reduce intestinal permeability and damage.[277]

Note, however, that whey protein contains glutamic acid (a.k.a. glutamate), *not* glutamine, though your body can make the latter out of the former.

Glutamine is required for leucine uptake into cells and the subsequent activation of protein synthesis. Glutamine supplementation has no effect on muscle mass or fat mass, but it may improve recovery from resistance training. It may also decrease the risk of falling ill from prolonged endurance exercise, notably by reducing exercise-induced dysfunctions of the intestinal tract.

Taurine

Taurine (L-taurine) is a sulfur-containing amino acid not involved in protein synthesis but omnipresent in your body. It is essential to cardiovascular function and the development and function of the brain, retina, and skeletal muscle.[278] Your body can make taurine from methionine and cysteine, so are there benefits to supplementation?

In young athletes, 1–6 grams of taurine improved endurance exercise performance regardless of how much taurine was taken or for how long.[279] This suggests that 1 gram is as effective as 6 and that chronic supplementation isn't necessary (just take your dose before your workout).

Taurine is also believed to benefit older adults with sarcopenia through its effects on protein metabolism, oxidative stress, and inflammation.[280] But this belief is based primarily on mechanistic evidence from studies in animals and test tubes, meaning the idea remains hypothetical until human studies are conducted.

Similarly, there are various levels of evidence that taurine supplementation may help with many other disease states, including neurodegenerative diseases, eye diseases, diabetes, heart failure, high blood pressure, and muscular dystrophy.[281] These conditions are associated with taurine depletion, so supplementation may help by restoring normal levels.

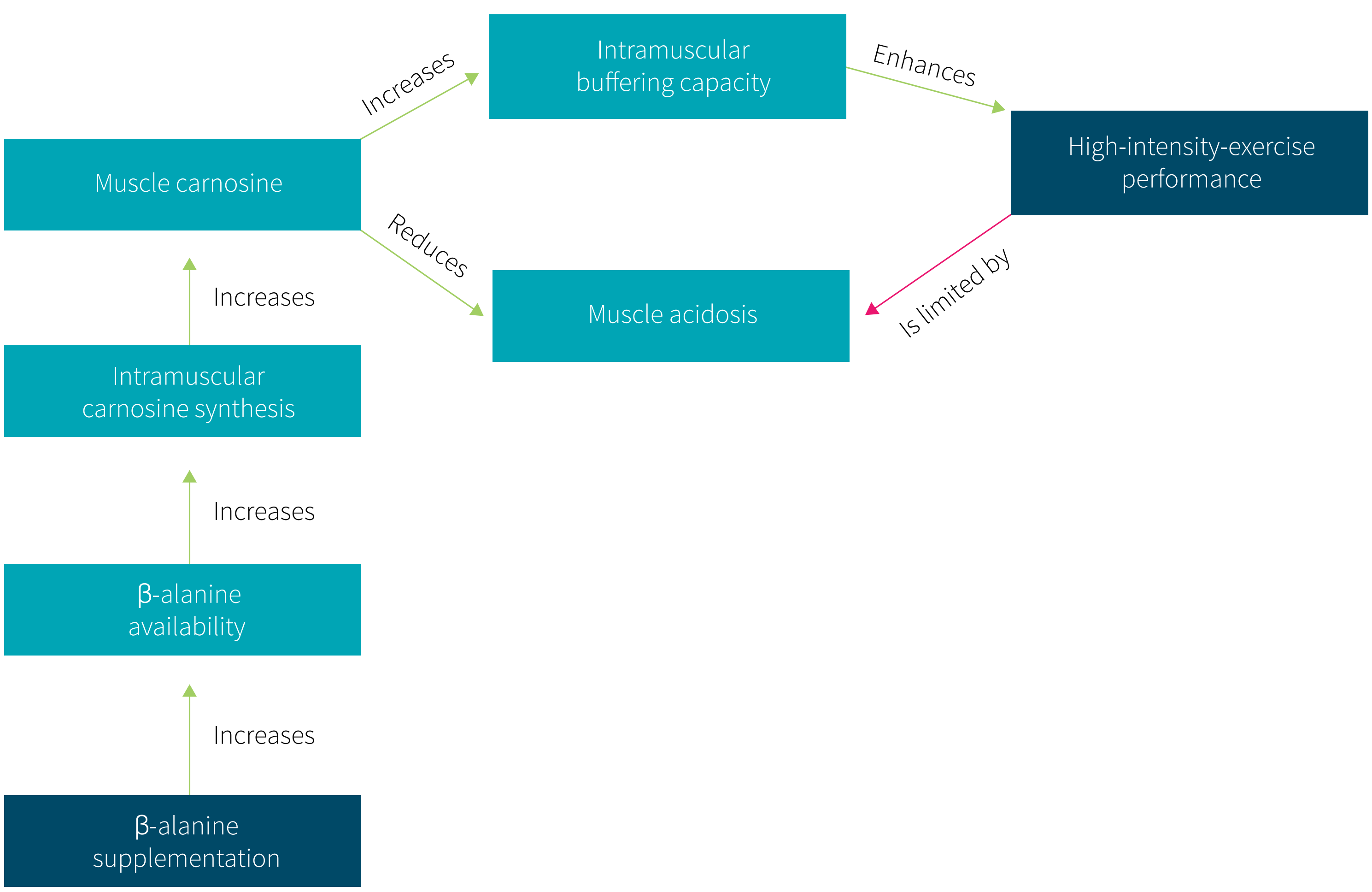
Taurine supplementation may improve endurance exercise performance and might benefit people with diseases of the cardiovascular, muscular, or nervous system.

β-alanine

Carnosine is synthesized in muscle tissues out of two amino acids: histidine and β-alanine. It helps buffer acids that form during muscular contractions.

You have more histidine available to make carnosine than you have β-alanine. In other words, β-alanine is the rate-limiting building block of carnosine.[282] There is thus good reason to believe that supplementing with β-alanine should increase carnitine concentrations in muscle tissues, and in such a way reduce fatigue and enhance athletic performance.

Figure 18: Effect of β-alanine on high-intensity-exercise performance.



β-alanine is one of the most heavily studied *ergogenic aids* (i.e., performance-enhancing supplements), and the results are promising. Chronic supplementation — taking β-alanine consistently for several weeks — increases endurance exercise performance during bouts of exercise lasting 1–10 minutes, regardless of fitness level or type of exercise.[283] Since the greatest benefit is for bouts of 1–3 minutes (with daily doses of 2.4–6.4 grams),[284] β-alanine should be an ideal supplement for combat-sport athletes such as boxers.[285]

Even the greatest benefit isn't large, mind you: a performance increase of some 2–3% on average. For most recreational athletes, such a minute advantage has little to no value. For competitors, though, it can make the difference between first and second place — or victory and defeat, in a boxing fight.

If your goal isn't to compete but to gain muscle or lose fat, however, you may want to spend your money on creatine and whey protein (and the *Fitness Guide*) rather than β-alanine, whose supplementation was shown to have no effect on body composition in collegiate football players and wrestlers,[286] untrained men beginning a resistance-training program,[287] and recreationally active women.[288]

Finally, let's address the elephant in the room: if the goal of β-alanine supplementation is to allow your muscles to synthesize more carnosine, then why not directly supplement carnosine? Because, in your blood, carnosine is rapidly broken down by the enzyme carnosinase, so that little to none reaches your muscle cells.[289] Instead, you find yourself with separate histidine and β-alanine; and since β-alanine, not histidine, is the rate-limiting building block of carnosine, taking β-alanine will increase your muscle's carnosine levels more than taking carnosine (as was demonstrated in mice).[290]

β-alanine doesn't affect body composition, but taking 2.4–6.4 grams daily for weeks increases endurance exercise performance during bouts lasting 1–10 minutes (with the greatest benefits seen for bouts of 1–3 minutes). Alas, the performance boost is so small as to be meaningful only for competitive athletes.

HMB

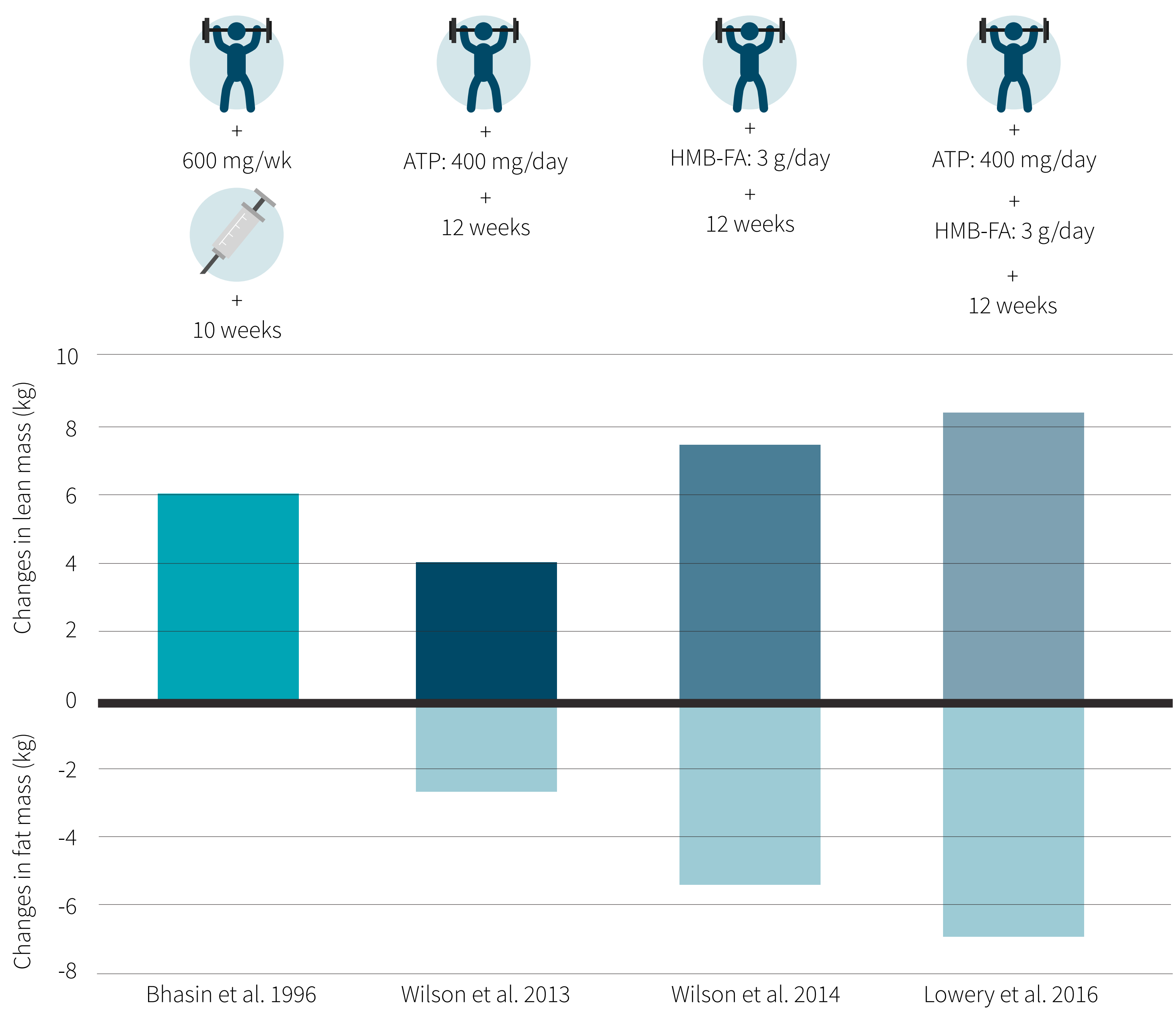
β-hydroxy-β-methylbutyric acid (HMB) is a metabolite of leucine with a slightly inferior ability to stimulate MPS and a superior ability to suppress MPB.[220,291] These effects appear to be similar between the two currently available forms, calcium HMB and HMB free acid (HMB-FA),[292] and have led to HMB being studied in muscle-wasting conditions,[293] such as cachexia and sarcopenia.[294,295]

HMB has also been investigated in resistance-trained populations, with some studies reporting massive increases in muscle mass (7.4–9 kilograms over 12 weeks) from 3 grams per day.[296,297,298] However, these findings have come under heavy criticism from experts in the field due to their implausibility and some shady data reporting.[299,300]

To put these numbers in perspective, consider that the data from 49 studies indicate that protein supplementation during a 12-week resistance-training program increases lean body mass by about 2.2 kilograms.[13] Even giving young men supraphysiological doses (doses greater than normally present in the body) of testosterone during a 10-week resistance-training program increased muscle mass by “only” 6.1 kilograms.[301]

Are we really going to believe that HMB (with purpoted muscle gains averaging 0.62–0.75 kg/wk) is more anabolic than an anabolic steroid (with muscle gains averaging 0.61 kg/wk)?

Figure 19: Changes in body composition from ATP, HMB-FA, and testosterone enanthate

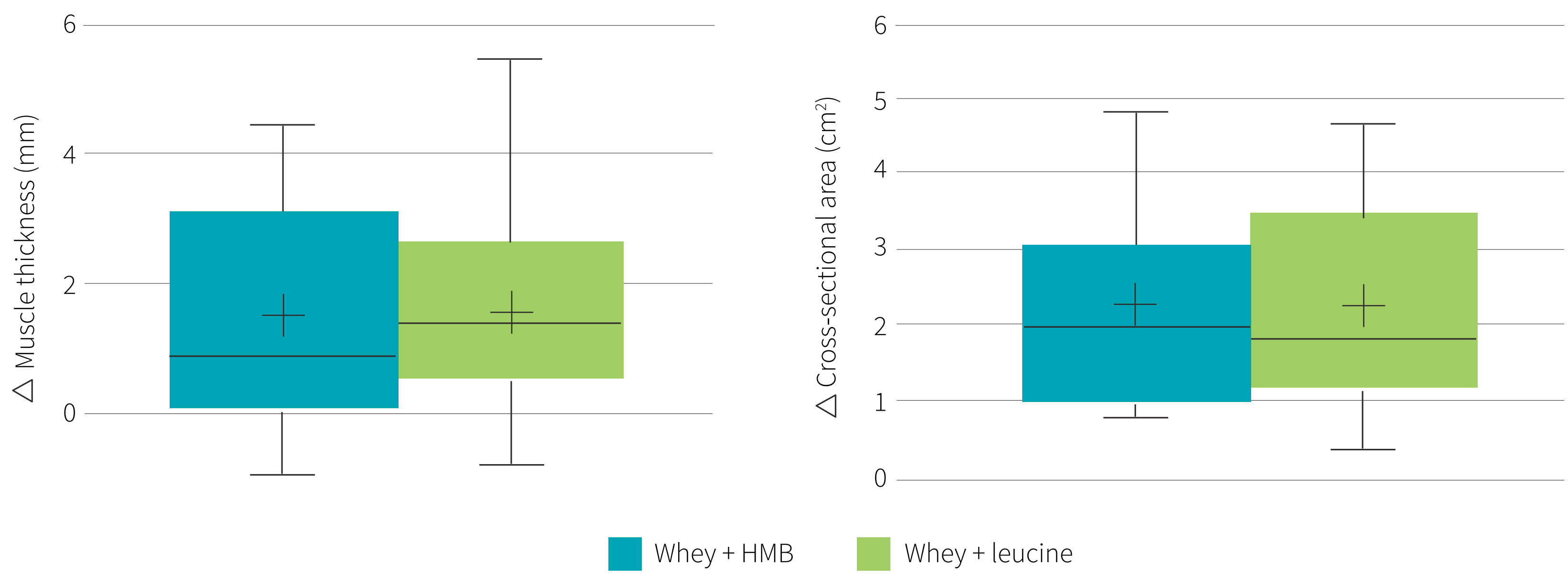


ATP: adenosine triphosphate

According to a meta-analysis of studies in competitive athletes and experienced weightlifters, 3 grams of HMB daily for 3–12 weeks doesn’t affect strength or body composition.[302] Yet a more recent study in competitive athletes (wrestlers, judokas, and practitioners of Brazilian jiu jitsu) did report that HMB increased lean mass (+1.5 kg) and reduced fat mass (−1.5 kg) over 12 weeks.[303]

Whether they reported a benefit or not, these studies pitted HMB supplementation against a placebo. When, instead, one group of resistance-trained men took 50 grams of whey protein plus 3 grams of HMB whereas the other took 50 grams of whey protein plus 3 grams of leucine, both groups experienced similar benefits in body composition and muscle size, thickness, and strength.[304]

Figure 20: Whey+HMB and whey+leucine lead to similar changes in muscle mass



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HMB does not appear to meaningfully affect the strength or body composition of athletes or resistance-training adults, but it can help with conditions involving muscle wasting, such as sarcopenia.

Amino acids as sources of nitric oxide

Arginine and citrulline (L-arginine and L-citrulline) are two amino acids commonly included in supplements marketed as increasing blood flow to the muscles, thereby helping with nutrient delivery, muscle growth, and exercise performance.

Arginine is the *nonessential amino acid* (NEAA) from which *nitric oxide* (NO) is made. However, in healthy individuals, supplementing with 6–10 grams of arginine doesn't appear to affect NO production,[305] blood flow to muscle tissue,[306] MPS,[307] or strength performance.[308]

Citrulline is usually sold as citrulline malate, a combination of L-citrulline and malic acid. It is a *nonprotein amino acid* (NPAA) converted into arginine in the kidneys, and supplementation does boost serum levels of both arginine and NO — but its effects on exercise performance are less clear.[309,310] Several studies in resistance-trained men and women reported that preworkout citrulline malate benefited weightlifting performance and, in the following days, reduced muscle soreness,[311,312,313,314] but studies in untrained or moderately trained adults reported no benefit.[315,316,317] Additionally, in resistance-trained men, chronic supplementation with L-citrulline or citrulline malate had no effect on body composition or muscle strength.[318]

Preworkout supplementation with citrulline might benefit weightlifting performance in trained adults, but its long-term effects on body composition and strength have been less investigated. Arginine supplementation doesn't affect nitric oxide production, blood flow, or muscle strength and shouldn't be used for these goals.

Lexicon

Amino acids (AAs). Organic compounds that participate in many functions in your body, from neurotransmission to the synthesis of enzymes, hormones, and of course proteins. All protein, including the protein you eat and the protein in your body, is made from some combination of twenty AAs.

Branched-chain amino acids (BCAAs). Leucine, isoleucine, and valine. The three BCAAs are considered the most anabolic of the nine EAAs and have therefore been marketed as a sports supplement. It is however possible that only leucine is especially anabolic, and that leucine taken alone is actually more anabolic than leucine taken with isoleucine and valine, due to competition for both absorption in the gut and entry into muscle tissue.

Endurance exercise performance. Time to exhaustion at a set pace. If you run longer/farther now than then, at the same speed, your endurance exercise performance has improved.

Ergogenic aid. Performance-enhancing supplement.

Essential amino acids (EAAs). Of the twenty AAs in protein, the nine your body cannot synthesize and thus needs to get from food

Fat-free mass. Another name for *lean mass*.

Fat mass. The fat content of your body.

Infusion. In a hospital setting, the continuous, slow introduction of a solution, usually into a vein. Intravenous infusions are commonly referred to as drips.

Lean mass. The nonfat content of your body, including your muscles, your bones, your organs minus their fat, and the water in your blood and cells.

Leucine. One of the three BCAAs and the most anabolic of the AAs.

Mass. A measure of how much matter is in an object. Unlike its weight, the mass of an object is constant. If an object's mass is 1 gram on Earth, its mass is 1 gram on the Moon.

Milk fat globule membrane (MFGM). A phospholipid-rich membrane under which the fat in milk is stored. Its consumption can increase muscle strength, neuromuscular efficiency, and physical function.

Muscle mass. The mass of your skeletal muscle. Not to be confused with your lean mass.

Muscle protein breakdown (MPB). The process of breaking down skeletal muscle tissue.

MPB is necessary for muscle growth, but for your muscle mass to increase, you need your MPS to exceed your MPB (overall, in the long term). Whether you exercise or not, however, your body is going to break down old or damaged muscle fibers to reuse what it can of their constituent AAs — to make new muscle fibers, enzymes, hormones, etc. When it comes to using AAs, MPS is among your body's lowest priorities; if your body needs AAs to serve as neurotransmitters, for instance, and you haven't eaten for a long time, it will scavenge even healthy muscle fibers.

Muscle protein synthesis (MPS). The process of building skeletal muscle tissue. Note that "protein synthesis" refers to the creation of any protein in your body; if your interest lies in muscle growth, you need to focus on MPS specifically.

Nonessential amino acids (NEAAs). Of the twenty AAs in protein, the eleven your body can synthesize.

Nonprotein amino acids (NPAAs). AAs that play a role in metabolism but are not involved in protein synthesis.

Weight. An object's relative mass. Unlike mass proper, weight is affected by gravity: it will be different on Earth and on the Moon; it can even vary on Earth (a given object is slightly heavier at sea level than at the top of a mountain, and at the equator than at the poles). However, for our purpose, mass and weight and pretty much interchangeable.

Whey protein concentrate (WPC). Whey from which most of the lactose, fat, bacteria, and other unwanted components have been filtered out, and which has then been dried. WPC is 29–89% protein. WPC used in the supplement industry is usually 80% protein.

Whey protein hydrolysate. A whey protein concentrate or isolate that has been "predigested" — meaning that the protein has been broken down into peptides (hydrolysates), primarily through enzymatic means.

Whey protein isolate (WPI). Whey from which most of the lactose, fat, bacteria, and other unwanted components have been filtered out, and which has then been dried. WPI is at least 90% protein.

List of abbreviations

α-MSH	α-melanocyte-stimulating hormone
AA	amino acid
Ace-K	acesulfame potassium
ADHD	attention-deficit hyperactivity disorder
ADI	Acceptable Daily Intake
AgRP	agouti-related protein
AMP	adenosine monophosphate
AMPK	AMP-activated protein kinase
ATP	adenosine triphosphate
BCAA	branched-chain amino acid
BCM	β-casomorphin
CCK	cholecystokinin
DIAAS	Digestible Indispensable Amino Acid Score
DOK7	docking protein 7
EAA	essential amino acid
EFSA	European Food Safety Authority
FDA	Food and Drug Administration
g	gram
g/kg/day	grams (of something consumed, usually) per kilogram of body weight
GLP-1	glucagon-like peptide-1
GMO	genetically modified
GMP	Good Manufacturing Practice **or** glycomacropeptide
GRAS	generally recognized as safe
HTST	high temperature, short time
HMB	β-hydroxy-β-methylbutyric acid
HMB-FA	HMB free acid
IAAO	Indicator Amino Acid Oxidation
ISSN	International Society of Sports Nutrition
kg	kilogram (1,000 grams)

lb	pound (≈ 454 grams)
m	meter
MFGM	milk fat globule membrane
mg	milligram (0.001 gram)
mm	millimeter (0.001 meter)
MPB	muscle protein breakdown
MPS	muscle protein synthesis
mTOR	mammalian target of rapamycin
NEAA	nonessential amino acid
nm	nanometer (0.000,000,001 meter)
NO	nitric oxide
NPAA	nonprotein amino acid
NPY	neuropeptide Y
PC	phosphatidylcholine
PDCAAS	Protein Digestibility-Corrected Amino Acid Score
PE	phosphatidylethanolamine
PhD	Doctor of Philosophy
PI	phosphatidylinositol
POMC	proopiomelanocortin
PYY	peptide YY
RDA	Recommended Dietary Allowance
wk	week
WPC	whey protein concentrate
WPI	whey protein isolate

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